

# AI for Agriculture: Methane Emissions Gap-Filling, Forecasting, and Scenario Analysis

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## **Abstract**

***Keywords***— keyword 1 - keyword 2 - keyword 3

## Acknowledgements

# Declaration

I, Nicholas Tobias, declare that the thesis has been composed by myself and that the work has not been submitted for any other degree or professional qualification. I confirm that the work submitted is my own, except where work which has formed part of jointly-authored publications has been included and referenced. The report may be freely copied and distributed provided the source is explicitly acknowledged.

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# Table of Contents

|                                                                             |            |
|-----------------------------------------------------------------------------|------------|
| <b>List of Figures</b>                                                      | <b>vii</b> |
| <b>List of Tables</b>                                                       | <b>xi</b>  |
| <b>1 Introduction</b>                                                       | <b>1</b>   |
| 1.1 Motivation . . . . .                                                    | 1          |
| 1.2 Research Questions . . . . .                                            | 2          |
| 1.3 Aims and Objectives . . . . .                                           | 2          |
| 1.4 Contributions . . . . .                                                 | 3          |
| 1.5 Report Overview . . . . .                                               | 4          |
| <b>2 Background and Related Work</b>                                        | <b>5</b>   |
| 2.1 North Wyke Farm Platform . . . . .                                      | 5          |
| 2.2 Eddy Covariance Measurement of FCH4 and Other Gases . . . . .           | 5          |
| 2.2.1 Gap-Filling Methodology and Its Difficulty . . . . .                  | 5          |
| 2.2.2 Lack of Forecasting/Scenario-Projection Work in This Sector . . . . . | 6          |
| 2.3 Towards Digital Twins and the What-If Simulation Gap . . . . .          | 7          |
| 2.4 Recent Developments in Tabular Modelling Approaches . . . . .           | 8          |
| 2.4.1 Tabular Foundation Models . . . . .                                   | 8          |
| 2.4.2 Sequence Architectures and the Attention-vs-Linear Debate . . . . .   | 8          |
| 2.5 Synthesis and Gap Statement . . . . .                                   | 9          |
| <b>3 Data</b>                                                               | <b>10</b>  |
| 3.1 Dataset . . . . .                                                       | 10         |
| 3.1.1 Site Description and Overview . . . . .                               | 10         |
| 3.1.2 Captured Metrics / Data Sources . . . . .                             | 10         |
| 3.2 Exploratory Data Analysis . . . . .                                     | 11         |

|          |                                                                  |           |
|----------|------------------------------------------------------------------|-----------|
| 3.2.1    | Availability of Data per Year . . . . .                          | 11        |
| 3.2.2    | Temporal Analysis and Dimensionality Reduction . . . . .         | 13        |
| 3.3      | Data Preprocessing Done . . . . .                                | 17        |
| 3.3.1    | Gap-Filling . . . . .                                            | 17        |
| 3.3.2    | Data Correction, Truncation, and Filtering . . . . .             | 17        |
| <b>4</b> | <b>Gap-Filling</b>                                               | <b>18</b> |
| 4.1      | Overview and Necessity of Gap-Filling . . . . .                  | 18        |
| 4.2      | Implementation Methodology . . . . .                             | 18        |
| 4.3      | Implementation Metrics and Evaluation Protocol . . . . .         | 19        |
| 4.4      | Replication of Baseline . . . . .                                | 21        |
| 4.4.1    | Overview of Models Used . . . . .                                | 21        |
| 4.4.2    | Overview of Features Used . . . . .                              | 21        |
| 4.4.3    | Results and Comparison with Cited Numbers . . . . .              | 21        |
| 4.5      | Improving the Baseline . . . . .                                 | 23        |
| 4.5.1    | Overview of Models Used . . . . .                                | 23        |
| 4.5.2    | Overview of Features Used . . . . .                              | 23        |
| 4.5.2.1  | Pooling Approach . . . . .                                       | 23        |
| 4.5.2.2  | Additional Features . . . . .                                    | 24        |
| 4.5.3    | Results and Comparison with Baseline . . . . .                   | 24        |
| 4.5.3.1  | Full Model Comparison . . . . .                                  | 24        |
| 4.5.3.2  | Robustness to Gap Length . . . . .                               | 27        |
| 4.5.3.3  | Daily Aggregation and Coverage Sensitivity . . . . .             | 28        |
| 4.5.3.4  | Feature, Pooling, and Refinement Ablations . . . . .             | 28        |
| 4.5.4    | Uncertainty Quantification for the Champion and Best Alternative | 29        |
| <b>5</b> | <b>Forecasting</b>                                               | <b>32</b> |
| 5.1      | Forecasting Methodology and Evaluation Metrics . . . . .         | 32        |

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| 5.1.1    | Baseline . . . . .                                          | 32        |
| 5.1.2    | Model Roster . . . . .                                      | 32        |
| 5.1.3    | Evaluation Metrics . . . . .                                | 33        |
| 5.2      | Training and Cross-Validation Methodology . . . . .         | 34        |
| 5.3      | Model Results . . . . .                                     | 34        |
| 5.3.1    | Full Observed-Target Model Comparison . . . . .             | 34        |
| 5.3.2    | Spike vs. Non-Spike Evaluation . . . . .                    | 39        |
| 5.3.3    | Evaluation Against the Gap-Filled Target . . . . .          | 43        |
| 5.3.4    | Interpretability . . . . .                                  | 43        |
| 5.3.5    | Uncertainty Quantification . . . . .                        | 47        |
| 5.3.5.1  | All-Model Uncertainty Comparison . . . . .                  | 47        |
| 5.3.5.2  | Conformal Calibration for Best Models . . . . .             | 48        |
| 5.3.5.3  | Spike-Coverage Correction (CQR) . . . . .                   | 50        |
| 5.3.5.4  | Livestock-Conditional Uncertainty . . . . .                 | 52        |
| <b>6</b> | <b>Scenario Projection</b>                                  | <b>55</b> |
| 6.1      | Outline of Scenario Projection Task . . . . .               | 55        |
| 6.2      | Interpretability and Feature Importance . . . . .           | 55        |
| 6.3      | Simulated Climate Dataset . . . . .                         | 56        |
| 6.4      | Data Pre-processing and Model Adjustment . . . . .          | 56        |
| 6.4.1    | Standardisation of New Data with the NWFP Dataset . . . . . | 56        |
| 6.4.2    | Generation of Simplified Scenario Model . . . . .           | 58        |
| 6.4.3    | Result of the Simplified Model . . . . .                    | 59        |
| 6.5      | Inference with Simulated Climate Dataset . . . . .          | 60        |
| 6.5.1    | Base Case . . . . .                                         | 60        |
| 6.5.2    | Induced Perturbations . . . . .                             | 60        |
| 6.5.2.1  | Livestock Density . . . . .                                 | 60        |
| 6.5.2.2  | Grazing Timing and Fertiliser Schedule . . . . .            | 66        |

|          |                                                                  |           |
|----------|------------------------------------------------------------------|-----------|
| 6.6      | Limitations . . . . .                                            | 68        |
| <b>7</b> | <b>Discussion</b>                                                | <b>70</b> |
| 7.1      | Main Findings . . . . .                                          | 70        |
| 7.1.1    | Overview of Model Performance . . . . .                          | 70        |
| 7.1.2    | Difficulty in Methane Flux Gap-Filling and Forecasting . . . . . | 71        |
| 7.2      | Limitations . . . . .                                            | 72        |
| 7.3      | Future Work . . . . .                                            | 73        |
| <b>8</b> | <b>Conclusion</b>                                                | <b>75</b> |
| 8.1      | Summary and Contribution . . . . .                               | 75        |
|          | <b>References</b>                                                | <b>78</b> |
|          | <b>Appendix A Source Code</b>                                    | <b>81</b> |

# List of Figures

|     |                                                                                                                                                                                                                                                      |    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 3.1 | Tower 2 EC CH <sub>4</sub> flux, daily median with 5th–95th percentile band – illustrating the spike-dominated character of the target and the years-long stretch of near-total absence following the 2019 land-use conversion. . . .                | 12 |
| 3.2 | Tower 2 FCH <sub>4</sub> data availability (% valid half-hours) by year and month, showing the 1,675-day gap from mid-2019 through the end of 2023. . . .                                                                                            | 12 |
| 3.3 | Distribution of FCH <sub>4</sub> gap lengths by tower. The 12-day (288-hour) dashed threshold marks the longest artificial-gap scenario used throughout the gap-filling evaluation protocol (Chapter 4). . . . .                                     | 13 |
| 3.4 | Hourly ACF and PACF of FCH <sub>4</sub> (raw gap-filled series) per tower, out to a 200-hour lag. The diurnal periodicity is visible as repeating peaks in the ACF; PACF decays sharply after the first few lags. . . . .                            | 15 |
| 3.5 | STL decomposition of hourly FCH <sub>4</sub> per tower (24-hour period): trend, daily-seasonal component (first week shown), and residual. . . . .                                                                                                   | 16 |
| 3.6 | D5 consensus feature-selection stability: % of cross-validation folds in which each meteorological variable was selected by the combined TICA/UMAP/t-SNE consensus ranking. . . . .                                                                  | 16 |
| 4.1 | MDS actual-vs-predicted FCH <sub>4</sub> , all five gap-length scenarios pooled, per tower – the literature-standard baseline’s near-zero/negative predictive skill is visible directly as scatter around the 1:1 line rather than along it. . . . . | 23 |
| 4.2 | Champion gap-filling check, Tower 4, 2021 (36% observed): daily-summed real observations against the gap-filled series produced by the improved pipeline. . . . .                                                                                    | 27 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4.3 | Prediction-interval coverage against the 90% nominal target (left) and interval-width sharpness against binned mean absolute error (right), QRF (RFm) vs. TabICLv2 quantile, all three towers, held-out gap-CV points. .                                                                                                                                                                                                                                                                                                   | 30 |
| 5.1 | Example 365-day recursive rollouts, all three towers, this dissertation's headline configuration (equal-weight blend of three TabPFN variants). Tower 4 and Tower 9 (anchored December 2022) visibly track the rise and fall of several spike clusters, not just the low-flux baseline – a clear qualitative improvement over this dissertation's earlier, un-refined TabPFN configuration. Tower 2 (anchored December 2018, its only anchor with real observed data, Section 3.2.1) is included for completeness. . . . . | 36 |
| 5.2 | p95 event-corrected TabPFN champion (BASE_ALL_52): permutation feature importance, pooled across all three towers and five rollout anchors. Livestock-related features (red) dominate the top of the ranking. . . . .                                                                                                                                                                                                                                                                                                      | 45 |
| 5.3 | Per-tower top-8 permutation importance. Tower 2's top features are exclusively meteorological, hydrological, and management-recency (wind speed, friction velocity, vapour-pressure deficit, catchment flow, fertiliser recency); Towers 4 and 9 both lead with livestock density and cattle density specifically. . . . .                                                                                                                                                                                                 | 46 |
| 5.4 | Species-disaggregated livestock-density importance: cattle density drives most of the species-split gain, though lamb density is a notable, unexplained exception (main text) and sheep density stays low. . . . .                                                                                                                                                                                                                                                                                                         | 46 |
| 5.5 | Fan chart, Tower 4, anchor December 2021, B18's TabPFN champion (BASE_ALL_52): median forecast and 90% conformal-calibrated interval against observed and gap-filled FCH4. The interval widens visibly during the mid-year, high-livestock-density window and still does not fully contain every large spike – the motivation for the CQR spike-coverage correction below. . . . .                                                                                                                                         | 50 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5.6 | Tower 4, anchor December 2021, B18's TabPFN champion: raw vs. CQR-corrected prediction intervals around observed and gap-filled FCH4, illustrating the selective widening that produces the spike-coverage gain in the table above. . . . .                                                                                                                                                                                     | 51 |
| 5.7 | Tower 2, anchor December 2018 (its only anchor with real observed data, Section 3.2.1): the model's raw, <i>uncalibrated</i> $[q_{05}, q_{95}]$ interval, shown hatched to distinguish it from the CQR-calibrated bands at Towers 4 and 9 above. No CQR margin exists for Tower 2 at any anchor – this is what "Tower 2 cannot support calibration" actually looks like, not a narrower version of the calibrated case. . . . . | 52 |
| 5.8 | Tower 4, anchor December 2021: flat CQR (lead-time-only margin) vs. livestock-density-stratified CQR intervals, with the underlying <b>fx_lsu_dens</b> trajectory shown below – the stratified interval visibly widens exactly where livestock density is high, independent of which specific model produced the underlying point forecast. . . . .                                                                             | 54 |
| 6.1 | Monthly-mean projected FCH4 to 2050 under the livestock-density ladder (SSP2-4.5, one representative GCM/realisation), all three towers, this chapter's current scenario engine. Own-historical-max visibly separates from the other three rungs at every tower. . . . .                                                                                                                                                        | 63 |
| 6.2 | Projected annual CH4 generation (kg CH4 per hectare per year) under the livestock-density ladder, SSP2-4.5, all three towers. Grazing-timing and fertiliser-schedule perturbations (Section 6.5.2) produce the analogous absolute-terms figures, not reproduced here since their percentage effects are already small (Table above, Section 6.5.2). . . . .                                                                     | 65 |

6.3 Monthly-mean projected FCH4 to 2050 under the grazing-timing shift  
(SSP2-4.5, one representative GCM/realisation), all three towers, this  
chapter’s current scenario engine. The shifted-timing curve sits visibly  
above the historical-timing curve at Towers 4 and 9, and is slightly below  
it at Tower 2. . . . . 67

# List of Tables

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1.1 | Research questions guiding this dissertation. . . . .                                                                                                                                                                                                                                                                                                                                                                                                             | 2  |
| 4.1 | Baseline replication results against published source figures. . . . .                                                                                                                                                                                                                                                                                                                                                                                            | 22 |
| 4.2 | Full gap-filling model comparison: median hourly $R_{OLS}^2$ (Zhu et al.’s convention, Section 4.3) across all five gap-length scenarios, all thirteen configurations tested. <sup>†</sup> TabPFN’s $R_{OLS}^2$ was not computed – deliberately scoped out, since its sklearn $R^2$ (0.459/0.401/0.402) already established it as a clear loser behind every tree-based or foundation-model result and did not warrant the additional recomputation cost. . . . . | 25 |
| 4.3 | $R_{OLS}^2$ improvement over the literature-standard MDS baseline (0.072/0.038/0.054), this chapter’s two headline results. . . . .                                                                                                                                                                                                                                                                                                                               | 26 |
| 4.4 | $R^2$ by gap-length scenario, each cell T2/T4/T9. . . . .                                                                                                                                                                                                                                                                                                                                                                                                         | 27 |
| 4.5 | Median $R^2$ under simple prediction averaging across model pairs/triples. . . . .                                                                                                                                                                                                                                                                                                                                                                                | 29 |
| 4.6 | Raw (uncalibrated) 90% prediction interval coverage and mean width by model and tower. . . . .                                                                                                                                                                                                                                                                                                                                                                    | 30 |
| 4.7 | Daily-resolution conformal calibration, RFm champion. . . . .                                                                                                                                                                                                                                                                                                                                                                                                     | 31 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5.1 | Recursive-rollout forecasting results, all-tower pooled, each model's best-selected configuration. $R^2$ is not reported for forecasting anywhere in this chapter (Section 5.1.3). <sup>†</sup> WAPE/Correlation for these two B18-refined rows were not part of the B18 refinement pass's own published metrics (Section 5.3.1, below); this dissertation computed them directly from the underlying saved prediction chains, using the same nine-tower-anchor-block evaluation set as every other B18 figure, and verified the computation bit-for-bit against the B18 pass's own published MAE and RMSE for both rows before reporting them here. The Seasonal Mean row is that same climatology baseline's own performance, evaluated on the identical nine-tower-anchor-block set (n=2,127) as the TabPFN and TabICLv2 rows above; its MASE is 1.000 by definition (a model matching the baseline exactly). Chain-persistence (the anchor day's real value held flat across the full 365-day rollout) is not used as a MASE reference anywhere in this dissertation (Section 5.1.3). . . . . | 37 |
| 5.2 | Spike-vs-non-spike performance, headline configuration's p95 event-corrected component (observed target, pooled across towers). . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 39 |
| 5.3 | Spike/non-spike split by tower, 90th-percentile threshold. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 40 |
| 5.4 | Pooled point-forecast performance, non-spike vs. spike days. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 41 |
| 5.5 | Baseline-scaled and uncertainty performance, non-spike vs. spike days. . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 41 |
| 5.6 | Tower-level non-spike / spike results. T2's 11 spike observations make its conditional estimates unstable; T9's near-zero spike correlation indicates essentially no ability to track spike-magnitude variation at that tower. . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 42 |
| 5.7 | The same model roster, scored against the gap-filled target instead of real observations – the ranking reverses, as flagged in Section 5.1.3. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 43 |

|      |                                                                                                                                                                                                                                                                                                                                      |    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5.8  | First-stage uncertainty comparison, T4/T9 pooled test rows (Tower 2 lacked sufficient calibration support). Raw quantiles as produced by each model, and the same after leave-one-anchor- out conformal calibration. . . . .                                                                                                         | 48 |
| 5.9  | Champion-focused conformal recalibration. <sup>‡</sup> <code>Direct_TabICLv2_solo_trend</code> on the restricted <code>FX_A_SPECIES</code> feature set – the actual scenario-projection engine used in Chapter 6, distinct from this chapter’s own TabICLv2 champion configuration. . . . .                                          | 48 |
| 5.10 | Spike-day interval coverage before and after CQR correction, both champion-tier models. . . . .                                                                                                                                                                                                                                      | 50 |
| 5.11 | Livestock-density-stratified CQR coverage, width, and pinball loss, both champion-tier models. . . . .                                                                                                                                                                                                                               | 53 |
| 6.1  | Per-GCM bias-correction factors applied to CMIP6 driver variables against NWFP’s own historical baseline. . . . .                                                                                                                                                                                                                    | 57 |
| 6.2  | Cost of simplifying to a scenario-realistic feature set, TabICLv2 only. $R^2$ is not reported for forecasting-style comparisons anywhere in this dissertation (Section 5.1.3). <sup>†</sup> Fertiliser-variable extension tested only against the superseded one-shot architecture, not retested against the current engine. . . . . | 59 |
| 6.3  | Annual-mean FCH4 change relative to the historically-observed baseline, under the bias-corrected livestock-density ladder, this chapter’s current <code>Direct_TabICLv2_solo_trend</code> scenario engine. . . . .                                                                                                                   | 61 |
| 6.4  | Projected annual CH4 generation (kg CH4 per hectare per year) under the livestock-density ladder, SSP2-4.5, this chapter’s current scenario engine. . . . .                                                                                                                                                                          | 64 |
| 6.5  | Annual-mean FCH4 change relative to baseline under grazing-timing and fertiliser perturbations (SSP2-4.5), this chapter’s current <code>Direct_TabICLv2_solo_trend</code> scenario engine. . . . .                                                                                                                                   | 66 |

# List of Abbreviations

ACF: Autocorrelation Function

ANN: Artificial Neural Network

AR: Autoregressive

CMIP6: Coupled Model Intercomparison Project Phase 6

CQR: Conformalized Quantile Regression

DL: Deep Learning

DOY: Day Of Year

EC: Eddy Covariance

FCH4: Methane Flux

FCO2: Carbon Dioxide Flux

GCM: Global Climate Model

GPP: Gross Primary Productivity

LSTM: Long Short-Term Memory

LSU: Livestock Unit

MASE: Mean Absolute Scaled Error

MDS: Marginal Distribution Sampling

MICE: Multivariate Imputation by Chained Equations

ML: Machine Learning

MPIW: Mean Prediction Interval Width

NWFP: North Wyke Farm Platform

PACF: Partial Autocorrelation Function

PCA: Principal Component Analysis

PICP: Prediction Interval Coverage Probability

QRF: Quantile Regression Forest

RF: Random Forest

RFm: Random Forest (met-driven variant)

SARIMAX: Seasonal AutoRegressive Integrated Moving Average with eXogenous regressors

SHAP: SHapley Additive exPlanations

SSP: Shared Socioeconomic Pathway

STL: Seasonal-Trend decomposition using Loess

SVM: Support Vector Machine

TFT: Temporal Fusion Transformer

TICA: Time-lagged Independent Component Analysis

UMAP: Uniform Manifold Approximation and Projection

UQ: Uncertainty Quantification

VPD: Vapour Pressure Deficit

VSN: Variable Selection Network

WAPE: Weighted Absolute Percentage Error

XGB: XGBoost (Extreme Gradient Boosting)

# 1 | Introduction

## 1.1 Motivation

Methane ( $\text{CH}_4$ ) is a powerful greenhouse gas, with a global warming potential roughly 28 times that of  $\text{CO}_2$  over a 100-year horizon and around 80 times that of  $\text{CO}_2$  over a 20-year horizon [1]. Agriculture is the leading sectoral source of anthropogenic methane, accounting for approximately 49% of agricultural greenhouse gas emissions through enteric fermentation alone in the UK and a comparably large share globally [2, 3], and global livestock emissions are projected to keep rising alongside demand for animal products [2]. Predicting  $\text{CH}_4$  flux at the ecosystem scale is genuinely difficult: emissions are driven by the continuous interaction of multiple, non-stationary biophysical factors – soil moisture, temperature, rainfall – together with discrete livestock management interventions that dominate the signal in UK managed grassland specifically.

The North Wyke Farm Platform (NWFP) offers a rare and underutilised resource to address this difficulty: continuous half-hourly Eddy Covariance (EC) flux measurements from managed beef cattle grassland spanning over seven years. To the author’s knowledge, no prior study has applied ML-based forecasting to EC  $\text{CH}_4$  flux at a managed temperate grassland – existing work in this space addresses gap-filling (within-distribution interpolation) rather than forecasting (sequential extrapolation), a structurally different modelling problem (Chapter 2). Closing this gap matters beyond academic interest: accurate prediction and scenario projection can support farm-level management decisions – grazing timing, stocking density – and inform scenario and policy planning toward national emissions-reduction targets.

This dissertation addresses the gap through a three-stage pipeline: gap-filling a continuous  $\text{CH}_4$  series from NWFP’s EC record (Chapter 4), forecasting that series using a benchmarked, uncertainty-quantified model roster (Chapter 5), and using the resulting

models to project CH<sub>4</sub> response under contrasting management and climate scenarios (Chapter 6).

## 1.2 Research Questions

| ID  | Question                                                                                                                                                                                                                    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RQ1 | What ML approaches exist for CH <sub>4</sub> flux prediction in agricultural/grassland contexts; do any address ecosystem-scale EC data; how does gap-filling differ from multi-step forecasting as a modelling constraint? |
| RQ2 | How do statistical baselines, tree-based ensembles, and deep learning compare for half-hourly EC CH <sub>4</sub> forecasting under temporal variability and non-stationarity?                                               |
| RQ3 | How transferable are findings from adjacent domains (wetland gap-filling, animal-scale prediction) to managed grassland EC forecasting?                                                                                     |
| RQ4 | How can XAI (SHAP) and UQ (quantile ML / conformal prediction) decode interactions between continuous environmental variables and discrete management interventions?                                                        |
| RQ5 | What structural requirements are needed to integrate forecasting models into a digital shadow with “what-if” scenario analysis at farm scale?                                                                               |

Table 1.1: Research questions guiding this dissertation.

## 1.3 Aims and Objectives

The overall aim of this dissertation is the development and evaluation of multiple machine learning approaches for CH<sub>4</sub> forecasting at NWFP, and to demonstrate their integration with scenario projection under contrasting management and climate conditions. This aim is pursued through five objectives:

1. Conduct a systematic literature review on time-series forecasting, quantile machine learning, eddy covariance sensor methodology, and agricultural digital twin frame-

works.

2. Acquire, pre-process, and document NWFP EC CH<sub>4</sub> data (2018–present), including gap imputation, quality control, and feature engineering.
3. Design and implement a benchmarking pipeline comparing statistical baselines, tree ensembles, and deep learning architectures, using temporal cross-validation to prevent data leakage.
4. Apply interpretability methods (SHAP) and uncertainty quantification (conformal prediction / quantile machine learning) to characterise model behaviour.
5. Generate synthetic management and climate scenarios and evaluate model behaviour under them.

## 1.4 Contributions

This dissertation contributes three benchmarked results to the field:

1. **A gap-filling benchmark for EC CH<sub>4</sub> flux at a managed grassland** (Chapter 4), comparing established methods (MDS, Random Forest), published baseline replications, and newer tabular/foundation model families, with uncertainty quantification and interpretability attached. The best configuration reaches an hourly-resolution  $R^2_{OLS}$  of 0.41–0.60 across all three towers (Zhu et al.’s own reporting convention, matched directly for comparability), against a published ceiling of  $R^2 < 0.1$  at hourly resolution for the same ecosystem type [4].
2. **A recursive multi-step forecasting benchmark for EC CH<sub>4</sub> flux** (Chapter 5), the first, to the author’s knowledge, applied to this ecosystem-scale signal at a managed temperate grassland, comparing statistical baselines, tree ensembles, and tabular foundation models under identical conditions, with calibrated uncertainty quantification. The best configuration – an equal-weight blend of three

TabPFN variants, including a conservative spike-excess correction – beats a day-of-year climatology baseline by a wide margin ( $\text{MASE} = 0.691$ ), though a dedicated spike-vs-non-spike evaluation shows this advantage is concentrated on ordinary days and essentially disappears on the high-magnitude events that dominate the  $\text{CH}_4$  budget (Chapter 5). Interpretability work across both gap-filling and forecasting converges on a single, repeatedly-confirmed finding: livestock density is the dominant driver of  $\text{CH}_4$  flux at NWFP, consistent across SHAP attribution, native feature importance, SARIMAX coefficients, and TabPFN permutation importance.

3. **Scenario projection under contrasting management and climate conditions** (Chapter 6): the forecasting architecture developed in this dissertation is extended to produce distinguishable, interpretable  $\text{CH}_4$  projections under synthetic livestock and climate scenarios, grounded in CMIP6-based climate data and externally-anchored, literature-plausible stocking levels.

## 1.5 Report Overview

Chapter 2 reviews the literature on EC flux gap-filling, forecasting, agricultural digital twins, and recent developments in tabular modelling, establishing the gap this dissertation addresses. Chapter 3 describes the NWFP dataset and the preprocessing pipeline built on top of it. Chapter 4 presents the gap-filling benchmark that produces a continuous  $\text{CH}_4$  series for training and evaluation, including its uncertainty quantification. Chapter 5 presents the recursive forecasting benchmark and its uncertainty quantification, this dissertation’s primary modelling contribution. Chapter 6 presents scenario projection under contrasting management and climate conditions. Chapter 7 discusses the findings and their limitations, and Chapter 8 concludes.

## 2 | Background and Related Work

### 2.1 North Wyke Farm Platform

The North Wyke Farm Platform (NWFP), operated by Rothamsted Research in Devon, UK, comprises three hydrologically isolated, intensively-monitored grazed grassland catchments – referred to throughout this dissertation as Towers 2, 4, and 9 – each instrumented with an Eddy Covariance (EC) flux tower since 2017–2018. The three catchments are managed as beef-cattle farmlets under a shared research programme, and NWFP’s broader research output, including EC-based flux studies of  $\text{CO}_2$  [5] and spatial variability in greenhouse gas fluxes at comparable UK sheep-grazed pastures [6], positions it as a long-running, representative case study of intensively-managed UK grassland livestock systems rather than a single, idiosyncratic site. NWFP has also been the subject of prior quantile-machine-learning work on soil moisture [7] and animal-scale methane prediction from the platform’s GreenFeed system [8], and is explicitly named as a candidate site for a farm-scale digital twin [9]. Full technical specifications of the dataset used in this dissertation – catchment areas, sensor configurations, and coordinates – are given in Chapter 3.

### 2.2 Eddy Covariance Measurement of $\text{FCH}_4$ and Other Gases

#### 2.2.1 Gap-Filling Methodology and Its Difficulty

The EC-ML literature for  $\text{CH}_4$  flux prediction is dominated by tree-based ensemble methods. Irvin et al. established a benchmark across 17 FLUXNET- $\text{CH}_4$  wetland sites, finding Random Forest and XGBoost superior to Artificial Neural Networks and penalised lin-

ear regression across all biomes, with soil temperature the most important predictor, and documented that raw ML uncertainty estimates were systematically underestimated across every site tested [10]. Kim et al. confirmed Random Forest dominance across five wetland and rice paddy sites, adding that lagged environmental variables improve  $\text{CH}_4$  prediction substantially more than  $\text{CO}_2$  does [11]. Zhu et al. extended these findings to challenging ecosystems including UK managed pastures, validating Random-Forest-based gap-filling over Marginal Distribution Sampling (MDS) as the recommended approach for precisely the ecosystem type studied at NWFP [4], and Kheradmand et al. confirm a similar pattern for  $\text{CO}_2$  gap-filling at grassland and cropland sites using tower-based EC data [12]. Across this literature, the models used are almost exclusively simple ANN and tree-based ensembles – a contrast this chapter returns to in Section 2.4.

### **2.2.2 Lack of Forecasting/Scenario-Projection Work in This Sector**

This literature is nonetheless answering a different question from the one this dissertation asks. Gap-filling is a within-distribution interpolation task: a model reconstructs missing observations inside a known regime, using both past and future context around the gap. Forecasting is sequential extrapolation: a model predicts forward from real or previously-predicted history alone, with no access to future context, and temporal dependencies become the dominant modelling challenge rather than a secondary concern. Every gap-filling benchmark reviewed above answers the former question; none addresses the latter. To the author’s knowledge, no prior study has applied ML-based forecasting to EC  $\text{CH}_4$  flux at a managed temperate grassland – this is the central gap this dissertation addresses.

## 2.3 Towards Digital Twins and the What-If Simulation Gap

Agricultural digital twins are difficult to build precisely because agricultural systems are living, weather-dependent, and non-stationary, unlike the controlled, deterministic environments industrial digital twins are usually built for. Purcell and Neubauer’s systematic review of agricultural digital twins finds that what-if scenario simulation – the capability that would transform a digital twin from a monitoring tool into a decision-support system – remains critically absent across the field [13]. Fakeye et al. propose a three-tier digital twin architecture specifically for NWFP and name CH<sub>4</sub> forecasting as a missing module within it [9]. Four components motivate this dissertation’s response to that gap:

1. **Forecasting** – as established in Section 2.2.2, this gap exists independently of any digital-twin framing.
2. **Scenario projection** – the specific what-if capability Purcell and Neubauer identify as missing.
3. **Interpretability** – Buzacott et al. demonstrate via Shapley-value analysis that raw predictive accuracy can mask a highly entangled multi-driver dependency in CH<sub>4</sub> emissions [14]; Sharma et al. caution that models lacking site-specific calibration frequently miss episodic, high-emission events [15]; Fuchs et al. show that ignoring discrete management interventions leads to systematic under-prediction of peak emission fluxes [16].
4. **Uncertainty quantification** – anchored on Irvin et al.’s finding that raw ML uncertainty is systematically underestimated [10]: a model that performs well on average but misrepresents its own confidence is not safe to use for scenario-based decision-support.

## 2.4 Recent Developments in Tabular Modelling Approaches

Section 2.2.1 shows the EC-ML gap-filling literature is dominated by simple ANN and tree-based ensembles. A newer generation of tabular and sequence architectures, not yet applied in this domain, motivates the model roster benchmarked in Chapters 4 and 5.

### 2.4.1 Tabular Foundation Models

TabPFN-family foundation models are pretrained on large volumes of synthetic tabular data and make predictions via in-context learning at inference time, without any dataset-specific parameter fitting. TabICL extends this approach to substantially larger tables via a two-stage column-then-row attention architecture [17], and TabICLv2 further improves generalisation and pretraining efficiency [18]. Hoo et al. extend the underlying TabPFN-v2 model to time-series forecasting by treating it as a tabular regression problem with lightweight temporal featurisation [19] – the same family of models this dissertation applies to CH<sub>4</sub> forecasting (Chapter 5) and scenario projection (Chapter 6), and a family with a documented structural limitation under genuine distribution shift, returned to in Chapter 6’s discussion of extrapolation risk.

### 2.4.2 Sequence Architectures and the Attention-vs-Linear Debate

Whether architectural sophistication translates into forecasting accuracy is itself an open question in the wider time-series literature. Zeng et al. demonstrate across nine standard long-term forecasting benchmarks that a set of embarrassingly simple linear models outperform sophisticated Transformer-based architectures, attributing the failure to permutation-invariant self-attention destroying temporal ordering information [20].

Montero-Manso and Hyndman further show that global forecasting methods – a single model fit across many related series – can match or exceed local, per-series methods without assuming similarity between series [21], directly relevant to this dissertation’s pooled and per-tower model comparisons. Together, this literature cautions against assuming any one architecture family will win in advance, motivating the identical-conditions benchmarking pipeline adopted throughout this dissertation.

## 2.5 Synthesis and Gap Statement

Read together, this literature establishes that ecosystem-scale  $\text{CH}_4$  forecasting at a managed temperate grassland is unaddressed: gap-filling benchmarks are mature but answer a structurally different question (Section 2.2); what-if scenario simulation is a named, open gap in the agricultural digital twin literature (Section 2.3); and the newer tabular and foundation model families capable of addressing both problems have not yet been applied in this domain (Section 2.4). To the author’s knowledge, no prior study has applied ML-based forecasting to EC  $\text{CH}_4$  flux at a managed temperate grassland. This dissertation’s specific combination – tabular and foundation models, applied to forecasting, scenario projection, interpretability, and uncertainty quantification together, at a representative UK managed grassland site – is the necessary and appropriate response to that gap.

### 3.1 Dataset

#### 3.1.1 Site Description and Overview

The North Wyke Farm Platform comprises three hydrologically isolated grazed-grassland catchments, each drained independently and instrumented with an Eddy Covariance flux tower: Tower 2, Tower 4, and Tower 9. Each catchment is farmed as a distinct beef-cattle system and is treated throughout this dissertation as an independent modelling unit – soil moisture, meteorological, and flux variables are never pooled across catchments without an explicit tower identifier, since doing so would mix physically distinct fields (Section 3.1.2). EC towers report half-hourly flux estimates; supporting meteorological and soil sensors report at 15-minute resolution; livestock location counts are recorded daily per catchment. Catchment area varies across the three sites and underlies the livestock-density (livestock-units per hectare) features used throughout Chapters 4 and 6. The EC record used in this dissertation spans 2018 to 2023 for training and evaluation, with 2024 reserved as a held-out window (Section 3.2.1).

#### 3.1.2 Captured Metrics / Data Sources

Three categories of source data are used as gap-filling features in this dissertation. EC flux and micrometeorological variables (30-minute resolution) provide the target (FCH4) alongside its established covariates: incoming shortwave radiation, air temperature, vapour pressure deficit, friction velocity (USTAR), wind speed, and soil heat flux. One naming detail is worth flagging as a concrete data-handling due-diligence point: shortwave radiation was initially expected under a `SW_IN_` column-naming pattern by analogy with standard FLUXNET conventions, but is recorded at NWFP under

SWIN\_1\_1\_1 instead – a reminder that every column mapping in this dataset was verified directly against the data rather than assumed from convention. Soil moisture and soil temperature (15-minute resolution) are recorded per catchment and are never averaged across towers (Section 3.1.1). Livestock location counts (daily, per catchment, per species) provide the livestock-density features identified as the dominant CH<sub>4</sub> driver in Chapters 4–6. Two further data sources exist in the raw NWFP export but were explored during EDA and not adopted as gap-filling features: 17-parameter catchment water-quality chemistry and raw animal body-condition scores, both retained in the raw data but outside this dissertation’s feature scope. Bodyweight, fertiliser-application recency, and catchment flow are used later in this dissertation (Chapters 5 and 6) and are introduced at that point rather than catalogued here, since they play no role in gap-filling itself.

## 3.2 Exploratory Data Analysis

Two properties of the raw data motivate the modelling pipeline that follows: FCH4 is severely incomplete (Section 3.2.1), and the dataset spans multiple measurement frequencies that must be reconciled onto a common temporal index before any model can be trained (Section 3.2.2).

### 3.2.1 Availability of Data per Year

FCH4 target completeness varies sharply across the three towers, computed directly from the consolidated hourly record: Tower 4 is missing 56.5% of hourly observations, Tower 9 is missing 75.0%, and Tower 2 is missing 88.5%. Tower 2’s incompleteness is dominated by a single, confirmed structural cause rather than instrument noise: a 1,675-day continuous gap from May 2019 to January 2024, corresponding to a genuine land-use change on the Red farmlet (field NW002) from permanent pasture to arable cultivation,

during which the EC tower’s flux footprint no longer represented grazed grassland. This gap is treated as a real missing-data mechanism, not something to be gap-filled through, and is returned to as a modelling limitation in Chapter 7. Separately, 2024 exists in the raw data but is deliberately withheld from all training and validation as the project’s final held-out evaluation window, consistent with the temporal train/test/hold-out split adopted throughout this dissertation.

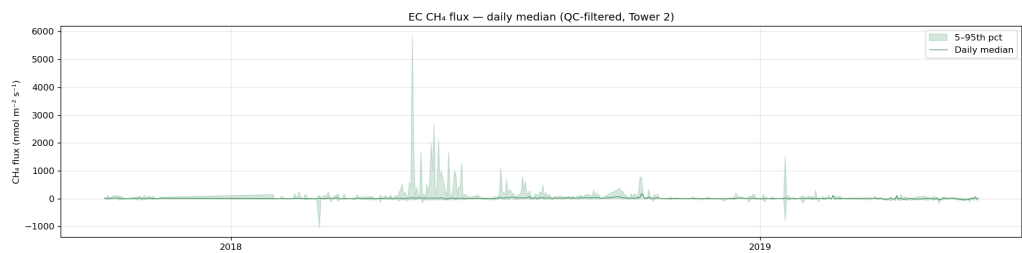


Figure 3.1: Tower 2 EC CH<sub>4</sub> flux, daily median with 5th–95th percentile band – illustrating the spike-dominated character of the target and the years-long stretch of near-total absence following the 2019 land-use conversion.



Figure 3.2: Tower 2 FCH<sub>4</sub> data availability (% valid half-hours) by year and month, showing the 1,675-day gap from mid-2019 through the end of 2023.

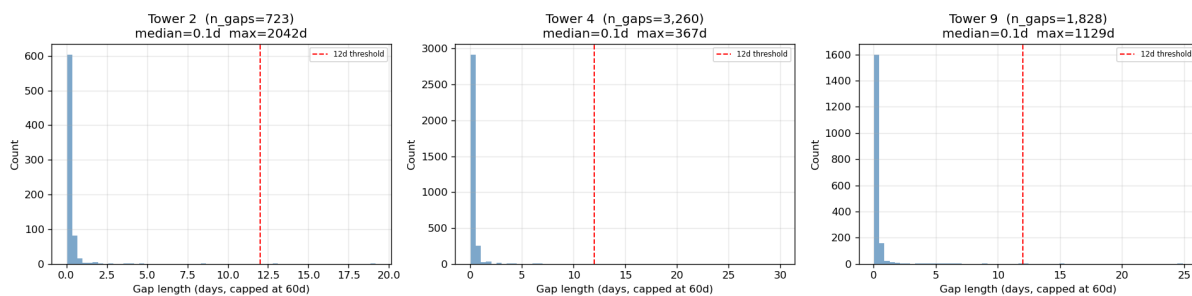


Figure 3.3: Distribution of FCH4 gap lengths by tower. The 12-day (288-hour) dashed threshold marks the longest artificial-gap scenario used throughout the gap-filling evaluation protocol (Chapter 4).

### 3.2.2 Temporal Analysis and Dimensionality Reduction

Beyond standard correlation and autocorrelation/partial-autocorrelation (ACF/PACF) diagnostics used throughout to characterise FCH4’s temporal dependence structure, two nonlinear dimensionality-reduction methods were applied to the environmental feature space: UMAP and Time-lagged Independent Component Analysis (TICA). TICA originates in molecular dynamics, where it identifies the slowest-evolving collective coordinates of a trajectory; unlike UMAP, which preserves static neighbourhood structure, TICA maximises autocorrelation over an explicit time lag, making it the more natural lens for a temporally-structured environmental record and the method preferred here. TICA was fitted per-tower on contiguous temporal segments at a 24-hour lag; UMAP and t-SNE embeddings were computed on a reproducible, tower-balanced sample (1,000 rows per tower), each coloured by tower identity, season, and FCH4 observed/missing status. A local 15-neighbour label-agreement test against chance found that towers separate well above chance (+0.13 to +0.18) and seasons separate even more strongly (+0.18 to +0.23), but FCH4-observed and FCH4-missing hours are barely distinguishable from chance (+0.04 to +0.05) – missing-target hours occupy essentially the same environmental neighbourhoods as observed ones. This is a reassuring result specifically for gap-filling: it supports interpolation from available training conditions rather than raising

a covariate-shift concern, since missingness in FCH4 does not correspond to a distinct region of environmental-feature space.

Building on this diagnostic, a fold-safe consensus feature-selection procedure was implemented using TICA loadings together with UMAP and t-SNE permutation importance, computed independently per outer cross-validation fold (30 folds: 3 towers  $\times$  5 scenarios  $\times$  2 repetitions) to avoid leakage, and combined into a single consensus ranking. This produced a remarkably stable result: a 6-variable core – soil temperature, soil moisture, air temperature, vapour pressure deficit, wind speed, and soil heat flux – was selected in every single fold, precipitation was excluded in every fold, and friction velocity (USTAR) was selected inconsistently (30% of folds). This consensus ranking underpins the feature sets used in the gap-filling benchmark of Chapter 4.

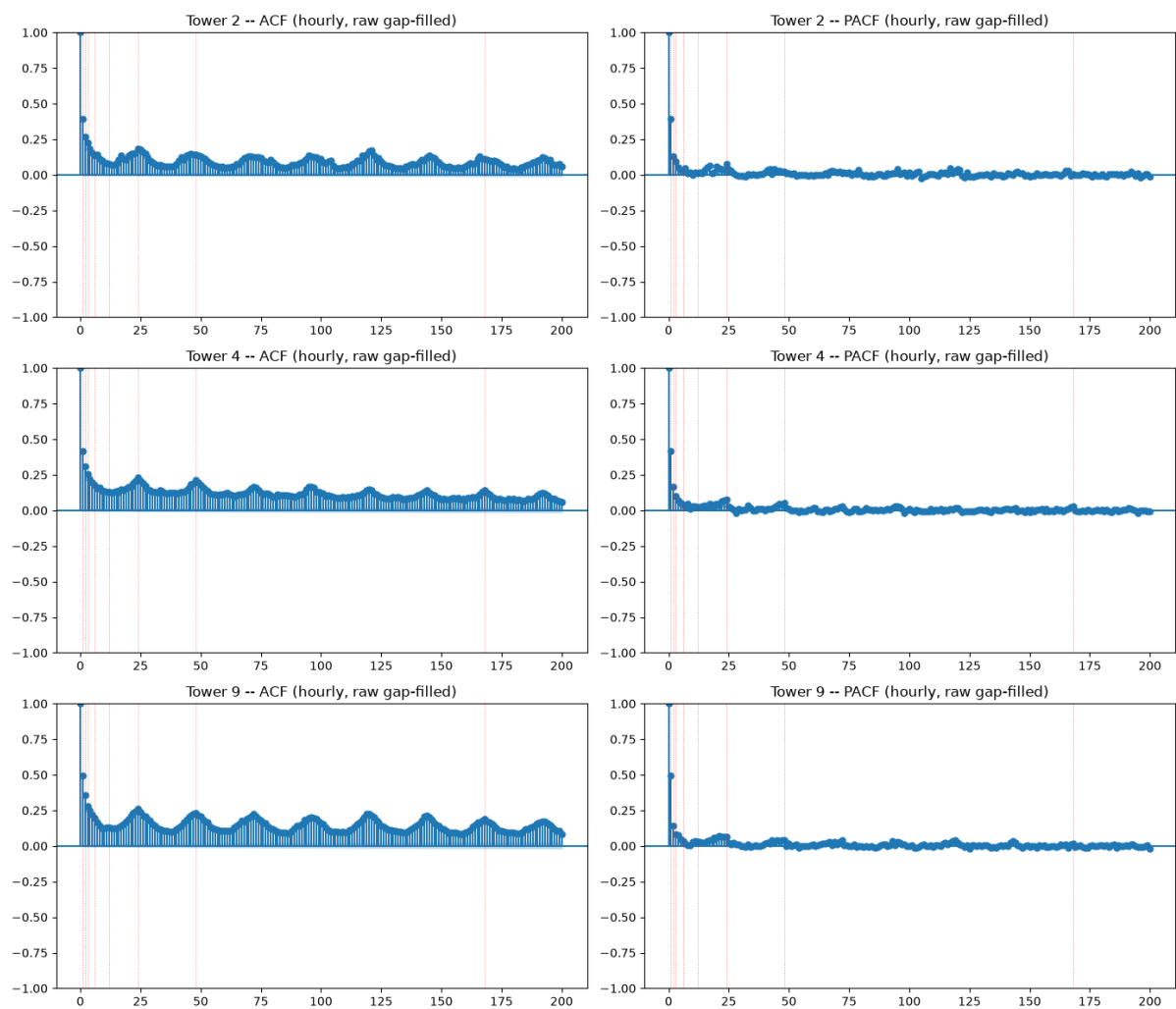


Figure 3.4: Hourly ACF and PACF of FCH4 (raw gap-filled series) per tower, out to a 200-hour lag. The diurnal periodicity is visible as repeating peaks in the ACF; PACF decays sharply after the first few lags.

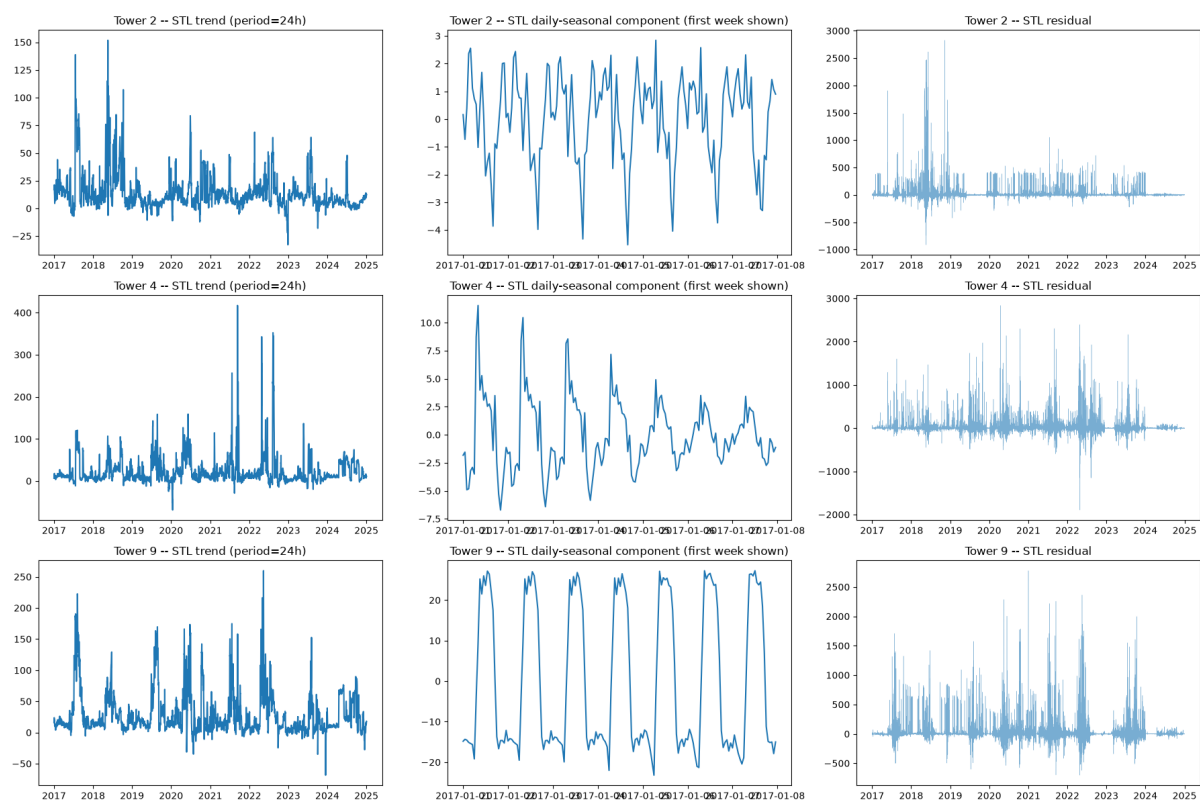


Figure 3.5: STL decomposition of hourly FCH4 per tower (24-hour period): trend, daily-seasonal component (first week shown), and residual.

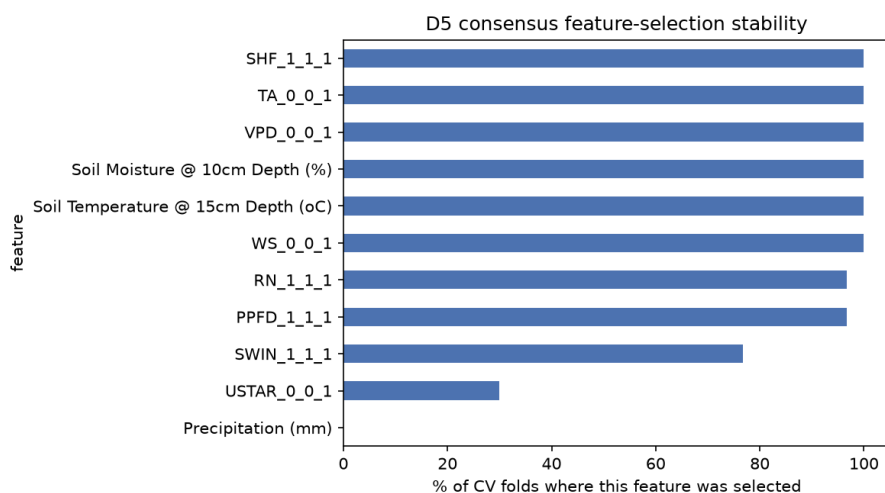


Figure 3.6: D5 consensus feature-selection stability: % of cross-validation folds in which each meteorological variable was selected by the combined TICA/UMAP/t-SNE consensus ranking.

## 3.3 Data Preprocessing Done

### 3.3.1 Gap-Filling

Meteorological driver variables (as distinct from the FCH4 target itself, whose gap-filling is this dissertation’s primary contribution and is presented in full in Chapter 4) are gap-filled using a Python reimplementation of REddyProc-style logic (`src/data/reddyproc_pipeline.py`, `mdc_gapfill()`), combining Marginal Distribution Sampling-style interpolation with a mean-diurnal-course fallback for extended gaps. This is a direct reimplementation of the standard EC gap-filling logic in Python rather than an invocation of the original R REddyProc package, which is not used anywhere in this project’s codebase.

### 3.3.2 Data Correction, Truncation, and Filtering

Two sensor-contamination issues were identified and corrected during preprocessing, both significant enough to be treated as data-correction events rather than routine filtering. First, a USTAR (friction velocity) contamination bug produced physically impossible readings as high as 1039.9 m/s, silently corrupting the mean-based gap-fill fallback during extended sensor blackouts. A matching contamination pattern was identified in vapour pressure deficit (VPD). Both were traced and corrected through a staged truncation fix applied to wind speed and air temperature inputs upstream of the affected derived variables. These corrections were applied before the dimensionality-reduction analysis of Section 3.2.3 and the gap-filling benchmark of Chapter 4, so results reported in both chapters reflect the corrected data throughout.

## 4.1 Overview and Necessity of Gap-Filling

FCH<sub>4</sub> at NWFP is severely incomplete (Section 3.2.1): 56.5% missing at Tower 4, 75.0% at Tower 9, and 88.5% at Tower 2. A continuous CH<sub>4</sub> series is a precondition for every downstream stage of this dissertation – forecasting (Chapter 5) requires an uninterrupted autoregressive history, and scenario projection (Chapter 6) requires a trustworthy baseline to project from – so gap-filling is not a preliminary data-cleaning step here but a modelling contribution in its own right, benchmarked, uncertainty-quantified, and evaluated against the published literature on its own terms. This chapter first replicates established gap-filling baselines from the literature under NWFP’s own conditions (Section 4.4), then improves on them using a wider model roster, richer features, and pooling across towers (Section 4.5), closing with the resulting champion’s uncertainty quantification.

## 4.2 Implementation Methodology

Thirteen configurations are benchmarked in this chapter, spanning from trivial statistical floors to tabular foundation models: a training-set mean (the trivial floor every other model must beat), Marginal Distribution Sampling (MDS, the literature standard), Random Forest with meteorological drivers only (RF, raw-driver baseline), MICE (multivariate imputation by chained equations), HyperImpute (an AutoML per-column imputer), the improved, feature-enriched Random Forest (RFm, this chapter’s pooled champion, Section 4.5), LightGBM, XGBoost, TabPFN, TabICLv2 (evaluated both pooled across towers and solo per tower), SAITS (a self-attention imputation network), and a bidirectional LSTM (Bi-LSTM). Two further models tested during this project’s development – SVM and ANN – appear only in the baseline-replication comparison (Section 4.4),

not in the improved-pipeline roster of Section 4.5, since ANN’s role there is specifically to replicate Kim et al.’s published comparison and SVM underperformed consistently across every baseline-replication scenario. TabICLv2 and TabPFN are the tabular foundation models introduced in Section 2.4 – TabICLv2 is used here in its regression (**TabICLRegressor**) configuration; it is evaluated in both a pooled and a solo-per-tower configuration, since its fixed 10,000-row context cap makes pooling actively harmful for this model specifically (unlike Random Forest, which benefits from pooling, Section 4.5.2) – the solo configuration is this chapter’s TabICLv2 headline result throughout. Uncertainty quantification is implemented for the two strongest models: quantile-regression-forest (QRF) intervals for RFm, and TabICLv2’s native quantile-regression output, both calibrated via conformal calibration (Section 4.5.4).

### 4.3 Implementation Metrics and Evaluation Protocol

The primary metric throughout this chapter is  $R^2$  at hourly resolution, alongside RMSE, MAE, and mean bias error (MBE). Every point metric reported in this chapter is the median across five artificial-gap scenarios of increasing length – 1 hour, 4 hours, 32 hours, 288 hours, and a mixed- length scenario – rather than a single fixed gap length, so that a model’s score reflects its robustness across realistic blackout durations rather than performance at one arbitrarily chosen gap length (gap-length-specific results are reported separately in Section 4.5.3.2). All models are evaluated under a blocked, calendar-gap cross-validation protocol with h-block buffering applied uniformly across all three towers: contiguous calendar blocks are held out as synthetic gaps, with a buffer window excluded from training immediately around each held-out block to prevent leakage from temporally adjacent observations. This mirrors the evaluation paradigm used by Irvin et al., Zhu et al., and Kim et al. (Chapter 2), and is the basis for direct comparison against their published numbers in Section 4.4.3.

One caveat applies throughout this chapter and is stated here rather than repeated at each comparison:  $R^2$  is not computed identically across the source papers being replicated, and this dissertation makes a deliberate, consistent choice between the two conventions in play. Scikit-learn’s residual-based  $R^2$  (`sklearn.metrics.r2_score`) is unbounded below and penalises systematic bias as well as scatter; Zhu et al. instead report a correlation-squared metric ( $R_{OLS}^2$ , equivalent to a squared Pearson correlation via `scipy.stats.linregress`), bounded to  $[0, 1]$  and insensitive to additive bias. **This dissertation reports  $R_{OLS}^2$  as its primary gap-filling metric throughout Section 4.5** – matching Zhu et al.’s own literature convention directly, and avoiding the interpretive burden of an unbounded-below statistic on a benchmark whose own literature ceiling is itself reported in  $R_{OLS}^2$  terms. The two conventions diverge sharply for MDS specifically (an early, uncorrected sklearn-based comparison found a 0.30 gap against  $R_{OLS}^2$  for MDS, before three implementation bugs in this project’s MDS reconstruction were corrected) but diverge only modestly for every Random Forest-, tree-, or TabICL-based model tested. Two scoped exceptions retain the sklearn convention: Section 4.4’s replication of Irvin et al., Zhu et al., and Kim et al.’s own published baseline figures necessarily preserves each source study’s original reporting convention for a faithful comparison; and the gap-length-robustness, daily-aggregation, and feature-ablation analyses later in Section 4.5.3 predate this chapter’s switch to  $R_{OLS}^2$  and are reported in their originally-computed sklearn  $R^2$ , since no  $R_{OLS}^2$  recomputation exists at that finer granularity – only Table 4.2’s headline comparison and the baseline-improvement analysis that follows it use  $R_{OLS}^2$ .

## 4.4 Replication of Baseline

### 4.4.1 Overview of Models Used

Three published gap-filling studies are replicated under NWFP’s own data and evaluation protocol: Irvin et al. (Random Forest and XGBoost against a 17-site FLUXNET-CH4 wetland benchmark), Zhu et al. (Random Forest against MDS, the specific comparison this dissertation’s improved pipeline is ultimately measured against, on UK managed pasture), and Kim et al. (Random Forest, ANN, and MDS, with an additional PCA-based dimensionality-reduction variant). Together these replications cover the model families – tree ensembles, MDS, and simple neural networks – that dominate the existing EC-CH4 gap-filling literature (Section 2.4).

### 4.4.2 Overview of Features Used

Each replication uses the feature set specified by its source paper as closely as NWFP’s own data allows: Irvin et al.’s replication uses meteorological drivers plus the co-measured EC fluxes (latent heat, sensible heat, FCO2) their paper includes; Zhu et al.’s replication deliberately excludes those co-measured fluxes, isolating a realistic meteorological-driver-only feature set (labelled RFm throughout this dissertation); Kim et al.’s replication adds lagged environmental variables and a PCA-reduced driver variant on top of the meteorological-only base.

### 4.4.3 Results and Comparison with Cited Numbers

Two findings replicate cleanly: Random Forest beats MDS at longer gap lengths (Zhu et al.’s central claim), and ANN becomes competitive with Random Forest at longer gap lengths (Kim et al.’s central claim, confirmed at Tower 4). One does not replicate: Kim et al.’s finding that PCA-reduced drivers degrade ML performance is not observed at

| Replication                             | Published result                                                                   | NWFP replication ( $R^2$ )                                                                                                                                                                                                    |
|-----------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Irvin et al. (RF/XGB, 17 wetland sites) | RF $R^2=0.79$ , XGB $R^2 \approx 0.65-0.67$                                        | T4: RF 0.144, XGB 0.086; T9: RF $-0.027$ , XGB $-0.089$                                                                                                                                                                       |
| Zhu et al. (RFm vs. MDS, UK pasture)    | RF beats MDS for gaps $>12$ days; both $R^2 < 0.1$                                 | T4: RFm $\approx -0.13$ , MDS $\approx -0.20$ ; T9: RFm $\approx -0.10$ (best), MDS $-0.58$ for 12-day gaps – RF-beats-MDS finding confirmed                                                                                  |
| Kim et al. (RF/ANN/MDS/PCA)             | RF best at short gaps, ANN competitive at longer gaps; PCA degrades ML performance | T4: RF best short (+0.136), ANN best medium/long/x-long (+0.091/+0.077/+0.057); T9: RF_lag best short-/medium (+0.152/+0.160), RF_PCA7 best long/xlong (+0.111/+0.056) – PCA-degrades-ML finding <b>not</b> confirmed at NWFP |

Table 4.1: Baseline replication results against published source figures.

NWFP – PCA-reduced Random Forest instead wins at longer gaps at Tower 9, a site-specific reversal. The magnitude gap against Irvin et al. (their wetland  $R^2$  of 0.79 vs. this dissertation’s 0.144 at best) is expected rather than a replication failure: Irvin et al.’s feature set includes co-measured EC fluxes strongly correlated with CH<sub>4</sub> at their sites, and NWFP’s own realistic driver-only feature set (Zhu et al.’s replication, directly above) is the fairer point of comparison for a deployable gap-filling pipeline – the premise the improved pipeline in Section 4.5 builds from. This early MDS/RFm replication uses Zhu et al.’s own exact feature scope and is kept distinct from the literature-corrected MDS and raw-driver-only RF floors reported in Section 4.5.3.1, which instead use this chapter’s own final, bug-fixed implementations.

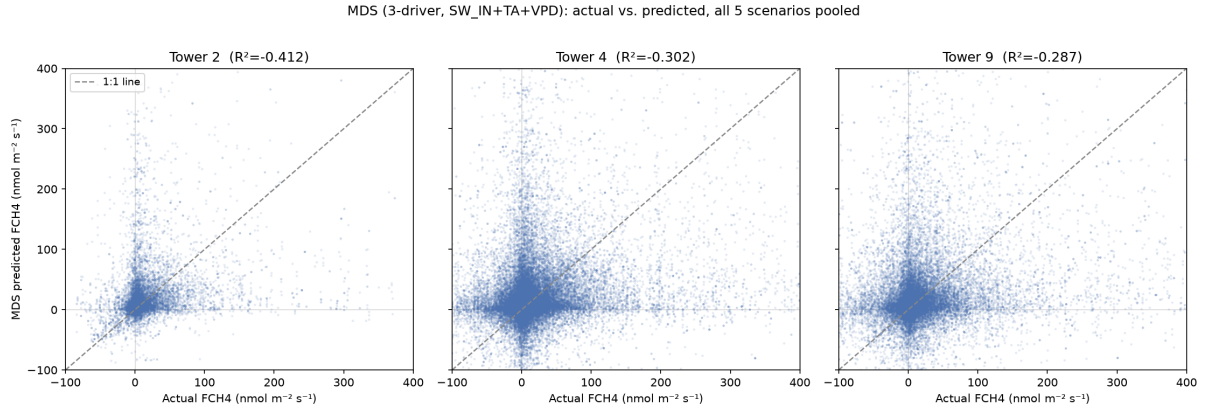


Figure 4.1: MDS actual-vs-predicted FCH4, all five gap-length scenarios pooled, per tower – the literature-standard baseline’s near-zero/negative predictive skill is visible directly as scatter around the 1:1 line rather than along it.

## 4.5 Improving the Baseline

### 4.5.1 Overview of Models Used

Building on the RFm baseline replicated above, the improved pipeline benchmarks the full roster introduced in Section 4.2 – from a trivial training-set-mean floor and the literature-standard MDS, through MICE and HyperImpute as general-purpose imputers, to RFm (this chapter’s pooled champion), LightGBM, XGBoost, TabPFN, TabICLv2 (pooled and solo), SAITS, and Bi-LSTM – against a common evaluation protocol (Section 4.3).

### 4.5.2 Overview of Features Used

#### 4.5.2.1 Pooling Approach

RFm is fitted as a partial pool across all three towers – a single model trained on T2+T4+T9 jointly, with tower-indicator dummy variables distinguishing catchments – rather than three independent per-tower models. This pooling is the single largest lever behind Tower 9’s improvement over the R-02 replication baseline (Section 4.4.3), since

it lets Tower 9’s shorter individual record borrow statistical strength from Towers 2 and 4’s longer records. TabICLv2 is the deliberate exception: it is evaluated both pooled and solo, and solo per-tower fitting is consistently the stronger configuration (Section 4.5.3.1), because its fixed context-row cap makes pooled context actively harmful rather than helpful for this specific architecture.

#### **4.5.2.2 Additional Features**

Beyond the meteorological drivers used in the RFm baseline replication, the improved pipeline adds: gap-filled FCO2 as a predictive feature (a proxy shown to substantially improve FCH4 prediction, Chapter 2); livestock stocking density (LSU/ha, this project’s single most important feature, confirmed independently across gap-filling, forecasting, and scenario projection, and returned to throughout this dissertation); pruned management-event recency features; gap-filled meteorological drivers and GPP/Reco via the REddyProc-style pipeline described in Section 3.3.1; and per-catchment external soil temperature. A dimensionality-reduction-informed consensus feature selection (Section 3.2.3) identified a stable 6-variable meteorological core (soil temperature, soil moisture, air temperature, VPD, wind speed, soil heat flux) underlying this feature set.

### **4.5.3 Results and Comparison with Baseline**

#### **4.5.3.1 Full Model Comparison**

Every row of the roster is reported here, not only the final champion and its closest challenger, to show the full build-up this dissertation’s result rests on: from a trivial floor and the literature- standard MDS baseline (both close to zero  $R_{OLS}^2$ , as Zhu et al. themselves report for this ecosystem type), through general-purpose imputers (MICE, HyperImpute) that recover real but limited signal, to the two models that lead this comparison. Solo-fitted TabICLv2 is this dissertation’s single best raw result, leading at

| Model or experiment                     | T2 $R_{OLS}^2$ | T4 $R_{OLS}^2$ | T9 $R_{OLS}^2$ |
|-----------------------------------------|----------------|----------------|----------------|
| Training-set mean                       | 0.000          | 0.000          | 0.000          |
| MDS                                     | 0.072          | 0.038          | 0.054          |
| RFm (meteorology only)                  | 0.054          | 0.042          | 0.065          |
| MICE (Bayesian ridge)                   | 0.104          | 0.128          | 0.110          |
| HyperImpute (AutoML)                    | 0.567          | 0.352          | 0.362          |
| <b>RFm, pooled (champion)</b>           | 0.601          | 0.409          | <b>0.430</b>   |
| LightGBM                                | 0.583          | 0.418          | 0.425          |
| XGBoost                                 | 0.619          | 0.380          | 0.375          |
| TabPFN <sup>†</sup>                     | –              | –              | –              |
| TabICLv2, pooled                        | 0.587          | 0.425          | 0.368          |
| SAITS                                   | 0.562          | 0.289          | 0.304          |
| Bidirectional LSTM                      | 0.280          | 0.204          | 0.189          |
| <b>TabICLv2, solo (best raw result)</b> | <b>0.686</b>   | <b>0.429</b>   | <b>0.430</b>   |

Table 4.2: Full gap-filling model comparison: median hourly  $R_{OLS}^2$  (Zhu et al.’s convention, Section 4.3) across all five gap-length scenarios, all thirteen configurations tested. <sup>†</sup>TabPFN’s  $R_{OLS}^2$  was not computed – deliberately scoped out, since its sklearn  $R^2$  (0.459/0.401/0.402) already established it as a clear loser behind every tree-based or foundation-model result and did not warrant the additional recomputation cost.

Tower 2 (+0.085 over RFm) and Tower 4 (+0.020), and matching RFm exactly at Tower 9 (0.430 for both, to three decimal places); RFm remains the stronger choice specifically where a matching production-grade uncertainty-quantification tool has been built around it (Section 4.5.4) – so both are reported as this chapter’s result, rather than declaring an unqualified single winner. Pooled TabICLv2 is included alongside its solo counterpart specifically to make this pooling effect visible: pooling costs TabICLv2 real accuracy at every tower (most severely at Tower 9, 0.368 vs. solo’s 0.430), confirming Section 4.5.2’s pooling caveat directly rather than asserting it. LightGBM and XGBoost, evaluated here as full roster members rather than only as a challenger footnote, each edge the champion at exactly one tower under  $R_{OLS}^2$  (LightGBM at Tower 4, +0.009; XGBoost at Tower 2, +0.018) but trail it at the other two; Bi-LSTM trails across the board, and SAITS – despite a substantial improvement over its earlier treatment in this project’s development (Section 4.5.4) – remains behind the champion at every tower too, though ahead of both deep-learning baselines.

One caveat applies to Tower 2’s comparatively strong  $R_{OLS}^2$  across every model in this section: it reflects a discriminable livestock-on/off signal specific to Tower 2’s land-use history (the 1,675-day permanent-pasture-to-arable conversion described in Section 3.2.1), not a genuinely easier underlying prediction task – a model at Tower 2 partly benefits from learning to distinguish two structurally different regimes rather than from resolving finer-grained CH<sub>4</sub>-flux variation within a single, stable regime the way Towers 4 and 9 require.

**Improvement over the baseline.** This project’s own methodological convention treats improvement over MDS, not absolute  $R^2$ , as the goal (a managed-grassland ecosystem is inherently open and noisy, so a high absolute ceiling was never expected). Table 4.2 makes that improvement directly measurable for the first time in  $R_{OLS}^2$  terms, the same units Zhu et al. themselves report MDS in:

| Improvement (MDS →)         | T2     | T4     | T9     |
|-----------------------------|--------|--------|--------|
| RFm champion                | +0.529 | +0.371 | +0.376 |
| TabICLv2 solo (best result) | +0.614 | +0.391 | +0.376 |

Table 4.3:  $R_{OLS}^2$  improvement over the literature-standard MDS baseline (0.072/0.038/0.054), this chapter’s two headline results.

Both the production-tooled RFm champion and the best-raw-accuracy TabICLv2 result improve on MDS by roughly 0.37–0.61  $R_{OLS}^2$  units at every tower – an eight-to-eleven-fold increase over MDS’s own score, not a marginal refinement. Tower 9, this project’s most consistently difficult site (Chapter 7), shows the smallest absolute gain in raw units (+0.376 either way) but still represents roughly an eight-fold improvement over MDS’s own 0.054 there. This is the improvement this dissertation is actually claiming: not a high absolute ceiling, which an open ecosystem-scale system was never going to offer, but a large, consistent, multiple-fold gain over the published literature standard at every tower.

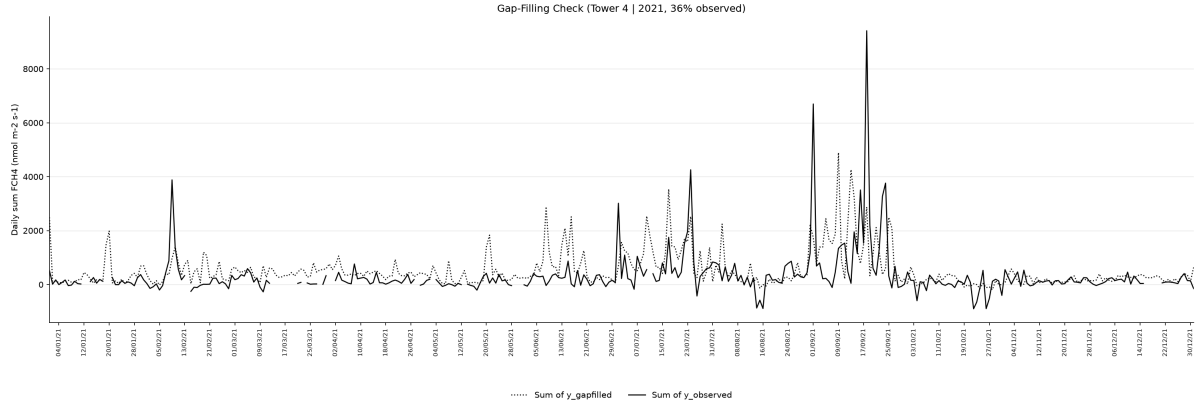


Figure 4.2: Champion gap-filling check, Tower 4, 2021 (36% observed): daily-summed real observations against the gap-filled series produced by the improved pipeline.

#### 4.5.3.2 Robustness to Gap Length

| Model              | 1 h            | 4 h            | 32 h           | 288 h            | mixed          |
|--------------------|----------------|----------------|----------------|------------------|----------------|
| LightGBM           | .522/.448/.447 | .663/.410/.406 | .646/.441/.422 | .033/.238/.177   | .470/.291/.422 |
| XGBoost            | .551/.389/.445 | .671/.349/.369 | .598/.363/.383 | -.647/-.062/.060 | .143/.228/.333 |
| TabPFN             | .459/.455/.443 | .704/.401/.402 | .684/.456/.543 | .092/.214/.190   | .456/.346/.350 |
| TabICLv2, pooled   | .558/.461/.459 | .722/.435/.364 | .698/.423/.467 | .218/.233/.231   | .466/.332/.330 |
| SAITS              | .510/.408/.378 | .219/.319/.285 | .659/.248/.338 | .055/.207/-.139  | .358/.293/.258 |
| Bidirectional LSTM | .423/.011/.146 | .237/.138/.150 | .546/.265/.167 | .148/.155/.072   | .189/.175/.112 |
| TabICLv2, solo     | .676/.521/.514 | .727/.465/.423 | .687/.428/.459 | .057/.206/-.102  | .428/.291/.227 |

Table 4.4:  $R^2$  by gap-length scenario, each cell T2/T4/T9.

The 288-hour scenario is the clearest weakness across the entire roster: several models, including the solo TabICLv2 champion at Tower 9, turn negative at this gap length even though their median-across-scenarios result (Table above) is comfortably positive. The headline  $R^2$  figures reported throughout this chapter – medians across all five scenarios – should accordingly not be read as evidence of equal reliability at every blackout duration; a 12-day (288-hour) blackout is this pipeline’s genuine stress case, for every model tested.

#### 4.5.3.3 Daily Aggregation and Coverage Sensitivity

Aggregating the champion RFm’s held-out hourly predictions to daily resolution raises  $R^2$  noticeably at every tower (unweighted full-scope: 0.698/0.504/0.485, vs. the hourly 0.576/0.404/ 0.426), as expected from averaging out hour-to-hour noise. This gain is sensitive to exactly how the aggregation is defined: restricting to days with at least 50% real hourly coverage lowers Tower 2 and Tower 4’s daily  $R^2$  (0.638/0.446) while raising Tower 9’s (0.586); weighting each day’s contribution by its number of observed hours raises all three slightly further (0.721/0.528/0.498). This sensitivity range is reported explicitly because a daily-resolution figure, quoted without its aggregation rule, can be made to look more or less favourable than the underlying hourly result supports – this dissertation’s primary reported metric throughout remains the hourly figure (Chapter 1), with the daily figures above reported only as a sensitivity check.

#### 4.5.3.4 Feature, Pooling, and Refinement Ablations

A wide range of feature and refinement variants were tested against the RFm/TabICLv2 champions; almost all were neutral or negative, and are recorded here for completeness rather than presented as adopted improvements. Replacing aggregate livestock density with species-disaggregated density changed RF  $R^2$  to 0.593/0.404/0.428 – a small Tower 2 gain, not a general improvement. Adding species and grazing-memory features together changed RF  $R^2$  to 0.601/0.395/0.425, a similar pattern: real at Tower 2, flat-to-negative elsewhere. A wide lag/lead feature expansion (164 features) degraded accuracy clearly (0.543/0.262/0.293 vs. the champion’s 0.576/0.404/0.426). Fertiliser-recency additions reduced gap-filling  $R^2$  for both RF and TabICLv2. TICA-derived components, tested as direct input features (distinct from their use as a feature-*selection* signal, Section 3.2.3), were essentially neutral; feeding a model’s own prediction-interval width back in as an input feature reduced RF’s  $R^2$  and severely damaged TabICLv2’s, particularly at Towers 2 and 9. TabICLv2 feature-dropping and TICA-component substitution were both flat

or negative. Native TabICLv2 hyperparameter variants stayed within  $-0.025$  to  $+0.019$   $R^2$  of the default configuration – a clean null result, nothing left on the table there. Row-cap bagging (fitting several TabICLv2 instances on different row subsets and averaging) improved only Tower 4, from 0.428 to approximately 0.440–0.441, plateauing by five to eight bags. Confidence-gated TabICLv2 self-training (promoting high-confidence predictions into the training pool as pseudo-labels) was neutral or harmful overall, its largest gain a marginal  $+0.005$   $R^2$  at Tower 4.

Simple prediction-averaging across the three strongest independent models was tested as a lower-effort alternative to further feature engineering:

| Tower | RF           | TabICLv2     | HyperImpute | RF + TabICLv2 | All three    |
|-------|--------------|--------------|-------------|---------------|--------------|
| T2    | 0.576        | <b>0.676</b> | 0.509       | 0.646         | 0.621        |
| T4    | 0.404        | 0.428        | 0.336       | <b>0.445</b>  | <b>0.445</b> |
| T9    | <b>0.426</b> | 0.423        | 0.354       | 0.432         | 0.417        |

Table 4.5: Median  $R^2$  under simple prediction averaging across model pairs/triples.

Averaging does not produce a general replacement for solo TabICLv2 – it loses at Tower 2, where TabICLv2 alone is already this chapter’s best individual result – but the Tower 4 gain (0.428 to 0.445) is real and reproducible, if tower-specific rather than a general-purpose improvement.

#### 4.5.4 Uncertainty Quantification for the Champion and Best Alternative

Both RFm and TabICLv2 have calibrated uncertainty intervals: quantile-regression-forest (QRF) intervals for RFm, and TabICLv2’s native quantile-regression output (evaluated in its pooled configuration, matching the models this uncertainty work was originally built against).

Raw coverage is already close to the 90% nominal target for both models at every tower

| Model               | Tower | $n$    | PICP (target 0.90) | MPIW  |
|---------------------|-------|--------|--------------------|-------|
| QRF (RFm)           | T2    | 12,384 | 0.941              | 162.7 |
| QRF (RFm)           | T4    | 48,707 | 0.919              | 197.1 |
| QRF (RFm)           | T9    | 28,189 | 0.898              | 213.7 |
| TabICLv2 (quantile) | T2    | 12,384 | 0.918              | 141.7 |
| TabICLv2 (quantile) | T4    | 48,707 | 0.901              | 184.2 |
| TabICLv2 (quantile) | T9    | 28,189 | 0.893              | 220.8 |

Table 4.6: Raw (uncalibrated) 90% prediction interval coverage and mean width by model and tower.

without any calibration step. Widths correlate moderately with absolute error (Pearson  $r = 0.48$ – $0.64$  across the six model/tower cells, strongest for QRF at Tower 2), indicating the raw interval width carries real, if imperfect, information about where each model is more or less likely to be wrong.

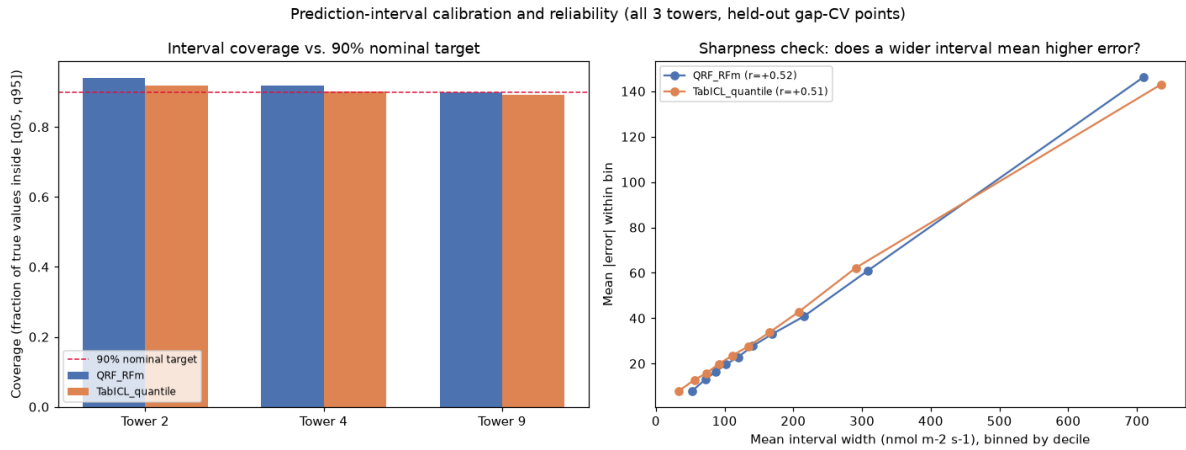


Figure 4.3: Prediction-interval coverage against the 90% nominal target (left) and interval-width sharpness against binned mean absolute error (right), QRF (RFm) vs. TabICLv2 quantile, all three towers, held-out gap-CV points.

Split-conformal calibration was applied on top of these raw intervals, moving coverage closer to 0.90 across individual tower/gap-scenario cells: calibrated QRF coverage ranged 0.876–0.912 and calibrated TabICLv2 coverage ranged 0.888–0.910 across the cells tested, with calibration narrowing already over-covering cells and widening under-covering ones, as intended. A separate daily- resolution conformal calibration was also tested:

| Tower | Calibration days | Test days | PICP  | MPIW  |
|-------|------------------|-----------|-------|-------|
| T2    | 112              | 112       | 0.866 | 104.7 |
| T4    | 430              | 430       | 0.923 | 161.5 |
| T9    | 245              | 245       | 0.861 | 124.7 |

Table 4.7: Daily-resolution conformal calibration, RFm champion.

TabICLv2 is ahead of QRF at two of three towers on the hourly comparison (Table 4.5) – sharper (narrower mean interval width) while sitting closer to nominal 90% coverage, rather than QRF’s more conservative, wider-than-necessary intervals at Towers 2 and 4. This is broadly consistent with TabICLv2 (pooled) also being a strong point-estimate model (Section 4.5.3.1), though the pooled configuration used for this uncertainty work is not the solo configuration that is this chapter’s overall best point estimate. One limitation applies to every interval reported in this section: this uncertainty-quantification work predates the solo-TabICLv2 point champion identified in Section 4.5.3.1, and interval width did not reliably grow with real (as opposed to synthetic) blackout length in the daily-resolution check above – these intervals should accordingly not yet be presented as fully calibrated uncertainty for the exact solo-TabICLv2 configuration this chapter otherwise recommends as its best point estimate.

## 5.1 Forecasting Methodology and Evaluation Metrics

Forecasting differs from gap-filling in a way that shapes every methodological choice in this chapter (Chapter 2): a forecaster has no access to future context, and predicts forward recursively from real or previously-predicted history alone. This dissertation evaluates forecasting exclusively via 365-day recursive rollout from real anchor points – feeding each model’s own prior-step prediction back in as input for the next step – rather than one-shot, fixed-horizon point forecasting, since recursive rollout is the setting a genuine forward-looking scenario tool (Chapter 6) actually requires.

### 5.1.1 Baseline

Two baselines anchor every result in this chapter: a day-of-year seasonal mean (the historical mean FCH4 for each calendar day,  $\pm 7$ -day window), and SARIMAX, a classical statistical time-series model fitted with exogenous meteorological and livestock regressors. seasonal mean is the baseline against which the primary metric, MASE, is scored throughout this dissertation (D-80, superseding an earlier chain-persistence convention on the grounds that holding one anchor-day value flat across a full year is not a baseline a practitioner would ever actually use, regardless of its own error magnitude); SARIMAX additionally serves as a competitive statistical model in its own right within the results below.

### 5.1.2 Model Roster

The tree-ensemble models introduced in Chapter 4 (Random Forest, XGBoost, LightGBM) and the tabular foundation models TabPFN and TabICLv2 are reused here in a sequential, recursive-rollout configuration rather than the tabular gap-filling config-

uration of Chapter 4 – the same underlying model families, but exercised in the fundamentally different forecasting setting described above. Two further architectures are introduced specifically for this chapter: the Temporal Fusion Transformer (TFT), an attention-based sequence model; DLinear, an embarrassingly simple linear decomposition forecaster motivated by Zeng et al.’s finding that such models can match or beat far more complex Transformer architectures on standard long-horizon benchmarks (Section 2.4); and a standard LSTM. An unweighted mean ensemble of Random Forest, XGBoost, LightGBM, and SARIMAX is also evaluated as a simple, low-effort combination strategy alongside the individual models.

### 5.1.3 Evaluation Metrics

MASE (Mean Absolute Scaled Error), scored against the day-of-year seasonal mean baseline above, is this dissertation’s primary forecasting metric throughout, alongside RMSE, WAPE, and Pearson correlation.  $R^2$  is deliberately not reported anywhere in this chapter’s forecasting results: on this spike-dominated, non-stationary target, a handful of extreme-error days can swing  $R^2$  by a large margin without reflecting any real change in forecast skill, making it an unreliable and non-representative metric for this specific task – a problem MASE, scored against a like-for-like seasonal baseline, does not share. (Gap-filling, evaluated on a fundamentally different, within-distribution interpolation task, does not share this problem and continues to report  $R^2$  throughout Chapter 4.) One scoring-convention caveat is stated here, before Section 5.3’s results, rather than after: every result in this chapter is scored against the real observed FCH4 target ( $y_{\text{observed}}$ ), not the gap-filled series produced in Chapter 4 ( $y_{\text{gapfilled}}$ ) that seeds each rollout’s autoregressive history. Scoring against  $y_{\text{gapfilled}}$  instead reverses the model ranking reported below (Section 5.3.3), because tree, SARIMAX, and ensemble models are trained by directly regressing onto that same gap-filled series and so track it circularly well, while TabPFN and TabICLv2 are evaluated zero-shot against  $y_{\text{observed}}$  only and receive no equivalent

boost.  $y_{\text{observed}}$  is the metric this dissertation treats as authoritative throughout, specifically because it is not inflated by this circularity.

## 5.2 Training and Cross-Validation Methodology

Every rollout result in this chapter uses a 5-anchor sweep – five distinct real calendar start-points, each rolled forward 365 days – rather than a single anchor, since single-anchor rankings proved unreliable in preliminary checks and do not appear as cited results anywhere in this chapter. Results are reported pooled across all three towers unless stated otherwise. An enriched feature set – species-disaggregated livestock density (cattle/sheep/lamb rather than a single combined stocking-density figure), a Tower-2 land-use flag, catchment flow, fertiliser- application recency, and liveweight density – was constructed uniformly across the full model roster. Tree, SARIMAX, and ensemble models showed no measurable gain from this enrichment in preliminary evaluation and were not pursued further given the additional compute cost, so their results in Section 5.3 use the unenriched base feature set (**BASE**); foundation and attention-based models (TabPFN, TabICLv2, TFT, LSTM, DLinear) were evaluated against several enriched configurations in full, and Table 5.1 reports each model’s own best-found configuration rather than a uniform one across the roster.

## 5.3 Model Results

### 5.3.1 Full Observed-Target Model Comparison

TabPFN leads this eleven-model comparison outright, in its standing headline configuration: an equal-weight blend of three TabPFN variants built on the same **BASE+ALL** enriched feature set (52 columns, including species-disaggregated livestock density) used throughout this chapter – the model’s own p95-quantile spike-excess correction, and two

further TabPFN fits trained on shorter, more recent history windows (1,095 and 1,460 days). Unlike the earlier, un-refined comparison (where the lowest-MASE model did not also lead on every other metric), the refined TabPFN row is the best entry in the table on all four columns shown – MAE, WAPE, and correlation, not only MASE. TabICLv2, refined the same way (a 1,460-day, tower-robust-scaled variant), moves up to a clear second place, ahead of every tree, ensemble, and statistical model. Both supersede this dissertation’s earlier, un-refined full-roster entries for these two architectures (TabPFN 0.7149, TabICLv2 0.7385) and beat the day-of-year seasonal-mean baseline by a wide margin. Among the remaining, un-refined entries, XGBoost has the lowest MAE (34.02) by a narrow margin over the two ensembles (34.06), and Random Forest the highest correlation (0.380) – neither changes the standing result, since MASE remains the metric this dissertation ranks models by throughout (Section 5.1.3). This refinement pass tested recent-history training windows, target normalisation, leakage-safe antecedent environmental/management features, spike-probability event correction, tower-adaptive model selection, and conservative blending; most did not help (richer antecedent features, tower-month normalisation, seasonal experts, and tower-adaptive model switching all made MASE worse or failed to generalise out of sample) – only the three ingredients above did, and only for TabPFN specifically (the TabICLv2 gain came from the recency window alone). The improvement over this dissertation’s prior best (a single, all-history direct TabPFN fit, MASE 0.6958) is not yet independently confirmed as statistically stable: a 10,000-resample block bootstrap over the available evaluation blocks gives a 95% interval for the MASE difference against that prior best which crosses zero ( $-0.0129$  to  $+0.0043$ ) – this dissertation reports the improvement plainly as its headline result while stating this caveat explicitly, rather than either suppressing the number or overstating its certainty. The best single, non-ensembled configuration – the p95 spike-excess correction alone, without the two-model blend – reaches MASE 0.6924, and is the configuration to cite where a single, interpretable model is preferred over the marginally sharper blend.

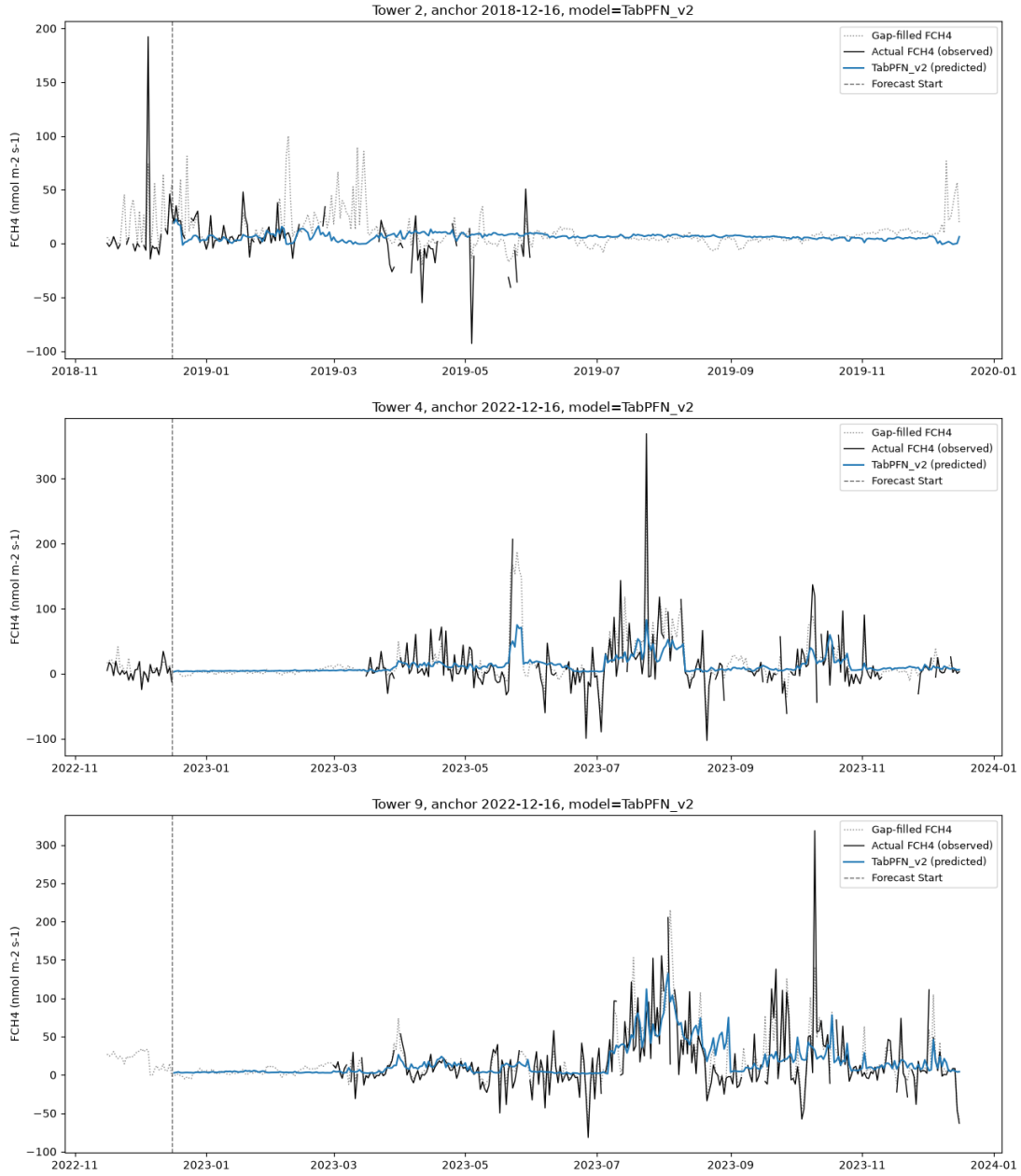


Figure 5.1: Example 365-day recursive rollouts, all three towers, this dissertation’s head-line configuration (equal-weight blend of three TabPFN variants). Tower 4 and Tower 9 (anchored December 2022) visibly track the rise and fall of several spike clusters, not just the low-flux baseline – a clear qualitative improvement over this dissertation’s earlier, un-refined TabPFN configuration. Tower 2 (anchored December 2018, its only anchor with real observed data, Section 3.2.1) is included for completeness.

| Model                           | Hyperparameters / configuration                                                                                                                               | MASE (seasonal mean) | MAE          | WAPE         | Correlation  |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------|--------------|--------------|
| <b>TabPFN</b>                   | Blend: p95+25% spike correction, 1,095d & 1,460d context windows                                                                                              | <b>0.6908</b>        | <b>29.82</b> | <b>0.797</b> | <b>0.491</b> |
| TabICLv2                        | 1,460-day context window, tower-robust target scaling                                                                                                         | 0.7191               | 30.86        | 0.825        | 0.443        |
| Ensemble                        | Equal-weighted mean of the RF/XGBoost/-LightGBM/SARIMAX rows below                                                                                            | 0.8020               | 34.06        | 0.991        | 0.379        |
| XGBoost                         | <code>max_depth=2</code> , <code>lr=0.02</code> , <code>n_estimators=400</code> , <code>min_child_weight=10</code> ; full 52-feature enriched set             | 0.8038               | 34.02        | 0.988        | 0.364        |
| TFT                             | <code>d_model=32</code> , 4 attention heads, <code>dropout=0.1</code> , 28-day lookback; full 52-feature enriched set                                         | 0.8120               | 35.13        | 1.009        | 0.236        |
| LightGBM                        | <code>num_leaves=7</code> , <code>min_child_samples=10</code> , <code>lr=0.02</code> , <code>n_estimators=400</code> ; driver/livestock-density features only | 0.8166               | 34.81        | 1.014        | 0.371        |
| Random Forest                   | <code>n_estimators=500</code> , <code>min_samples_leaf=10</code> , <code>max_features=0.5</code> ; driver/livestock-density features only                     | 0.8411               | 35.17        | 1.045        | 0.380        |
| SARIMAX                         | AIC-selected order ( $p \in \{1, 2\}$ , $d = 1$ , $q \in \{0, 1\}$ per tower/anchor); driver/livestock-density features only                                  | 0.8741               | 36.11        | 1.082        | 0.348        |
| LSTM                            | Hidden size 64, 28-day lookback; + species-disaggregated livestock density                                                                                    | 0.9521               | 40.22        | 1.154        | 0.275        |
| DLinear                         | Trend/seasonal decomposition, <code>kernel=25</code> , 28-day lookback; + bodyweight density                                                                  | 1.1872               | 45.74        | 1.519        | 0.277        |
| <i>Seasonal Mean (baseline)</i> | <i>Day-of-year mean, <math>\pm 7</math>-day circular window, computed from real pre-anchor history only</i>                                                   | <i>1.000</i>         | 43.79        | 1.170        | -0.004       |

Table 5.1: Recursive-rollout forecasting results, all-tower pooled, each model’s best-selected configuration.  $R^2$  is not reported for forecasting anywhere in this chapter (Section 5.1.3). <sup>†</sup>WAPE/Correlation for these two B18-refined rows were not part of the B18 refinement pass’s own published metrics (Section 5.3.1, below); this dissertation computed them directly from the underlying saved prediction chains, using the same nine-tower-anchor-block evaluation set as every other B18 figure, and verified the computation bit-for-bit against the B18 pass’s own published MAE and RMSE for both rows before reporting them here. The Seasonal Mean row is that same climatology baseline’s own performance, evaluated on the identical nine-tower-anchor-block set ( $n=2,127$ ) as the TabPFN and TabICLv2 rows above; its MASE is 1.000 by definition (a model matching the baseline exactly). Chain-persistence (the anchor day’s real value held flat across the full 365-day rollout) is not used as a MASE reference anywhere in this dissertation (Section 5.1.3).

Tellingly, this same refinement pass confirms rather than resolves this chapter’s central weakness: scored on pre-anchor-estimated spike days specifically, the p95 event-corrected component’s MASE rises to 3.28, against 0.52 on non-spike days – the spike-blindness problem quantified in full below persists even after this targeted correction attempt.

DLinear and LSTM are decisively the weakest configurations tested, both performing far worse than every tree, statistical, or foundation model in this comparison – directly contradicting the "simple linear models beat complex architectures" finding Zeng et al. report on standard benchmarks (Section 2.4): on this dissertation’s spike-dominated, non-stationary target, model simplicity alone was not sufficient. TabICLv2’s point-estimate result above reflects a corrected extraction of its predicted median (the 0.5 quantile column) rather than its predicted mean, which is materially biased high on this heavy-tailed, spike-dominated flux distribution; all TabICLv2 numbers in this chapter use the corrected, median-based extraction.

A hyperparameter-tuning pass on LightGBM (rollout-specific tuning: reduced leaf count, minimum child samples, and learning rate) found a real single-tower improvement that did not generalise: the tuned configuration improved Tower 4’s accuracy, but reusing the identical tuned hyperparameters at Tower 9 produced a clear regression rather than a comparable gain – a negative result retained here specifically because it illustrates that hyperparameter tuning validated on one tower cannot be assumed to transfer to another, reinforcing this dissertation’s standing convention (Chapter 1) of always reporting full three-tower coverage rather than a single-tower result. Separately, in this dissertation’s original, un-refined full-roster screen, TabPFN with species-disaggregated features was effectively tied with a bodyweight-density variant (MASE=0.7150 vs. 0.7149) – species-disaggregation is the configuration this dissertation’s interpretability work is built around (Section 5.3.4, Chapter 6), and remains so under the refined headline configuration, whose **BASE+ALL** feature set includes species-disaggregated density as a subset.

### 5.3.2 Spike vs. Non-Spike Evaluation

Table 5.1’s aggregate MASE is pooled across every day of every rollout, and this masks a sharp, practically important split in where that headline advantage actually comes from. This section evaluates the standing headline configuration directly, then uses a finer-grained breakdown from this dissertation’s earlier, un-refined TabPFN configuration as a deep-dive supplement, since it covers every tower individually and the headline configuration’s own check confirms the identical pattern at a coarser grain.

**The standing headline configuration.** Using tower-specific 95th-percentile thresholds estimated only from pre-anchor observations (a stricter cut than the 90th-percentile threshold used in the deep-dive below), the p95 event-corrected component of this dissertation’s headline blend splits into 1,997 non-spike and 130 spike observed evaluation points, pooled across towers:

| Metric                          | Non-spike ( $n = 1,997$ ) | Spike ( $n = 130$ ) |
|---------------------------------|---------------------------|---------------------|
| MASE                            | <b>0.524</b>              | <b>3.285</b>        |
| MAE                             | 20.31                     | 176.48              |
| RMSE                            | 30.01                     | 210.12              |
| Bias (prediction – observation) | +1.48                     | –176.48             |

Table 5.2: Spike-vs-non-spike performance, headline configuration’s p95 event-corrected component (observed target, pooled across towers).

The pattern is unambiguous: MASE is a genuinely strong 0.524 on ordinary days but degrades to 3.285 on spike days – over six times worse, and well above the seasonal-mean baseline (MASE= 1) rather than approaching it. This figure is not directly comparable in magnitude to the deep-dive’s own spike-day MASE below (which uses a looser 90th-percentile threshold covering a broader, less extreme set of events) – the qualitative conclusion, not the exact multiple, is what carries across both thresholds: spike days remain this configuration’s dominant failure mode. Bias is almost entirely a spike-

day phenomenon (average  $-176.5 \text{ nmol m}^{-2} \text{ s}^{-1}$  on spikes, against a negligible  $+1.5$  on non-spike days): the refinement pass that produced this dissertation’s headline result improved typical-day accuracy and partially corrected spike magnitude, but did not close the underlying gap. The remainder of this section uses the earlier configuration’s per-tower saved daily chains to go further than this pooled check allows – percentile sensitivity, per-tower breakdown, and spike-event detection – while carrying the caveat above that its exact figures predate the refinement.

**Deep-dive (earlier, un-refined configuration).** This detailed breakdown evaluates the champion configuration (TabPFN, BASE+species) separately on spike days – defined retrospectively as an observed daily FCH4 value at or above each tower’s own 90th percentile, the same definition used for the conditional-coverage check in Section 5.3.5 – and on the remaining, non-spike days. This is a diagnostic stratification on observed outcomes, not a deployable spike detector, evaluated on the saved daily median forecast chains (BASE+species, all three towers, anchors 2018–2022).

| Tower        | 90th-percentile threshold | Non-spike $n$ | Spike $n$  |
|--------------|---------------------------|---------------|------------|
| T2           | 23.25                     | 91            | 11         |
| T4           | 86.39                     | 1,188         | 132        |
| T9           | 112.53                    | 810           | 90         |
| <b>Total</b> | tower-specific            | <b>2,089</b>  | <b>233</b> |

Table 5.3: Spike/non-spike split by tower, 90th-percentile threshold.

The mean spike prediction captures only 17.3% of the mean observed spike magnitude, and every single one of the 233 spike observations in this evaluation is underpredicted. WAPE looks marginally better on spikes only because its denominator grows with observed magnitude – absolute error is in fact roughly eight times larger than on non-spike days.

This table is the central finding of this deep-dive: this earlier configuration’s own sea-

| Metric                          | Non-spike | Spike         |
|---------------------------------|-----------|---------------|
| Observed mean                   | 14.41     | <b>188.84</b> |
| Predicted mean                  | 14.84     | <b>32.69</b>  |
| Bias (prediction – observation) | +0.43     | –156.15       |
| MAE                             | 19.38     | <b>156.15</b> |
| RMSE                            | 27.91     | <b>191.03</b> |
| Correlation                     | 0.329     | <b>0.131</b>  |
| WAPE                            | 0.869     | 0.827         |
| Underprediction rate            | 46.6%     | <b>100.0%</b> |

Table 5.4: Pooled point-forecast performance, non-spike vs. spike days.

| Metric                       | Non-spike    | Spike        |
|------------------------------|--------------|--------------|
| seasonal mean MASE           | <b>0.659</b> | <b>1.001</b> |
| seasonal mean RMSSE          | 0.699        | 0.995        |
| Persistence MASE             | 0.911        | 0.909        |
| Native 90% interval coverage | 84.3%        | <b>50.2%</b> |
| Mean native interval width   | 107.15       | 183.75       |
| Mean native pinball loss     | 6.32         | <b>49.95</b> |

Table 5.5: Baseline-scaled and uncertainty performance, non-spike vs. spike days.

sonal mean-scaled MASE advantage (0.7150 pooled, no longer Table 5.1’s top entry but still this dissertation’s most thoroughly spike-audited configuration) is concentrated almost entirely on non-spike days (MASE 0.659, a genuine 34% improvement over seasonal mean). On spike days the model is effectively tied with the seasonal mean (MASE  $\approx 1.001$ ) – the model’s advantage over a naive seasonal baseline essentially disappears exactly on the days that matter most for the CH<sub>4</sub> budget, the same qualitative conclusion the headline configuration’s own coarser check reaches above. Native 90% interval coverage falls from 84.3% to 50.2% on spikes even though mean interval width widens by about 72%, and mean pinball loss rises almost eightfold; Conformalized Quantile Regression, applied to this same champion configuration, improves spike coverage to approximately 57% – a real improvement over the raw figure above, but still well short of the nominal 90% target.

A direct *spike-detection* check – flagging a predicted spike whenever the median forecast

| Tower | MAE            | Correlation          | seasonal mean MASE   | Native 90% coverage |
|-------|----------------|----------------------|----------------------|---------------------|
| T2    | 11.53 / 27.06  | 0.052 / 0.106        | 0.357 / <b>1.357</b> | 0.802 / 0.818       |
| T4    | 17.11 / 155.49 | 0.350 / 0.143        | 0.700 / <b>0.990</b> | 0.905 / 0.530       |
| T9    | 23.59 / 172.91 | 0.312 / <b>0.001</b> | 0.625 / <b>0.962</b> | 0.758 / 0.422       |

Table 5.6: Tower-level non-spike / spike results. T2’s 11 spike observations make its conditional estimates unstable; T9’s near-zero spike correlation indicates essentially no ability to track spike-magnitude variation at that tower.

itself exceeds the same tower-specific threshold used to label observed spikes – confirms the model is not merely imprecise on spike magnitude but structurally blind to spike *occurrence*: of 233 real spikes, only 1 is flagged as a true positive (recall 0.43%,  $F1 = 0.008$ ), against 2,086 true negatives (specificity 99.86%). The model’s apparent reliability comes entirely from correctly predicting the much larger population of non-spike days, not from any genuine event-detection skill. This conclusion is not an artefact of the 90th-percentile threshold choice: raising the definition to the top 5% (P95) sharpens it further – observed spike magnitude rises from a mean of 128.64 (P80) to 257.33 (P95) while the mean prediction stays essentially flat at 30–35 throughout, and native coverage degrades monotonically from 62.4% to 35.9%.

Taken together, this section supports a narrower reading of this dissertation’s headline forecasting result than the pooled MASE alone suggests, for both the deep-dive configuration and the current standing headline alike. TabPFN-based models reduce typical-day absolute error relative to the seasonal mean and are nearly unbiased across the bottom 90% of the observed distribution, but do not reliably reproduce, detect, or quantify methane spikes: on spike days their advantage relative to the seasonal mean shrinks sharply or disappears, every spike in the deep-dive sample is underpredicted, and calibrated uncertainty intervals undercover by roughly 40 percentage points. This is the central limitation of this dissertation’s forecasting contribution and is returned to explicitly in Chapter 7.

### 5.3.3 Evaluation Against the Gap-Filled Target

| Model                    | MASE (gap-filled) | RMSE         | WAPE         | Correlation  |
|--------------------------|-------------------|--------------|--------------|--------------|
| Ensemble (MASE-weighted) | <b>0.666</b>      | <b>25.38</b> | <b>0.704</b> | 0.522        |
| Ensemble (unweighted)    | 0.666             | 25.39        | 0.705        | 0.523        |
| XGBoost                  | 0.677             | 25.87        | 0.725        | 0.477        |
| LightGBM                 | 0.689             | 26.11        | 0.728        | 0.496        |
| Random Forest            | 0.703             | 25.85        | 0.756        | <b>0.529</b> |
| SARIMAX                  | 0.831             | 29.21        | 0.919        | 0.463        |
| TabPFN                   | 0.903             | 34.90        | 0.814        | 0.240        |
| TabICLv2                 | 0.922             | 35.27        | 0.841        | 0.210        |
| TFT                      | 1.021             | 36.10        | 1.103        | 0.147        |
| LSTM                     | 1.122             | 39.79        | 1.188        | 0.185        |
| DLinear                  | 1.502             | 42.59        | 1.653        | 0.182        |

Table 5.7: The same model roster, scored against the gap-filled target instead of real observations – the ranking reverses, as flagged in Section 5.1.3.

Scoring against the gap-filled target reverses Table 5.1’s ranking almost exactly: the ensemble, tree, and SARIMAX models now lead, and TabPFN/TabICLv2 fall to the middle of the pack. This is not evidence that the gap-filled target is a more truthful evaluation – it demonstrates that locally fitted ensembles track the smoother, gap-filled reconstruction they are trained to regress onto more closely, whereas the zero-shot foundation models see no equivalent benefit, exactly the circularity risk flagged in Section 5.1.3. Table 5.1’s observed-target ranking remains this dissertation’s authoritative comparison throughout.

### 5.3.4 Interpretability

Interpretability analysis converges on a single, repeatedly-confirmed finding across every method capable of producing feature attributions on this model roster: livestock density is the dominant driver of FCH4 at NWFP. This holds under native tree-model feature importance and SHAP attribution (Random Forest, XGBoost, LightGBM), SARIMAX regression coefficients, and TabPFN permutation importance, at every tower except Tower

2. SHAP importance for livestock density grows with forecast lead time rather than shrinking, from a mean  $|\text{SHAP}|$  of 7.6 at 1–7 days to 38.4 at 181–270 days (Tower 4) – livestock presence becomes a stronger, not weaker, predictor the further into the future a forecast reaches.

The permutation-importance pass below targets this chapter’s headline blend’s own single-model component specifically – the p95 event-corrected direct-regression TabPFN champion (BASE\_ALL\_52, MASE 0.6924, Section 5.3.1) – rather than the earlier, now-superseded TS-wrapper TabPFN+species configuration. `fx_lsu_dens` (combined livestock density) again leads the overall ranking by a wide margin (mean importance 1.746), but species-disaggregation now places a bodyweight-density feature absent from the earlier configuration’s own feature set – `fx_total_liveweight_dens` – directly behind it (1.652), with `fx_cattle_dens` close behind in third (1.550): cattle-driven livestock density remains a clear top-3 signal, though no longer literally "the" second-ranked feature once this richer feature set’s own bodyweight-density term is included. Grazing activity, wind speed, and friction velocity round out the next several ranks (`fx_grazing_active` 0.595, `fx_WS_mean` 0.498, `fx_USTAR_mean` 0.314). One genuine new wrinkle emerges under this architecture that was not present in the earlier pass: `fx_lamb_dens` jumps to 8th overall (0.280), against near-bottom in the earlier configuration, while `fx_sheep_dens` stays low (0.034) – this dissertation flags it as a plausible artefact of the pooled, tower-dummy-encoded architecture (which can pick up cross-tower sensitivity patterns a per-tower time-series wrapper cannot) rather than a substantive new finding, and it does not change the headline cattle-dominance conclusion.

Per-tower, Towers 4 and 9 both reproduce the identical top-3 ordering seen overall (Tower 4: `fx_lsu_dens` 2.864, `fx_total_liveweight_dens` 2.618, `fx_cattle_dens` 2.433; Tower 9: 2.372, 2.334, 2.216 respectively) – overwhelming livestock dominance at both. Tower 2’s own top-10, by contrast, contains zero livestock features under this architecture either, the same finding as every prior interpretability pass on this project, though its specific

composition is meteorological, hydrological, and management-recency rather than purely thermal: wind speed and friction velocity lead (`fx_WS_mean`, `fx_USTAR_mean`), followed by vapour-pressure deficit, catchment flow (7- and 14-day rolling means), fertiliser application rate and recency, shortwave radiation, lagged soil moisture, and day-of-year phase – consistent with Tower 2’s own muted livestock signal established in Section 4.5.3 and returned to in Chapter 6, even though the exact meteorological drivers involved differ from the earlier configuration’s own top-10.

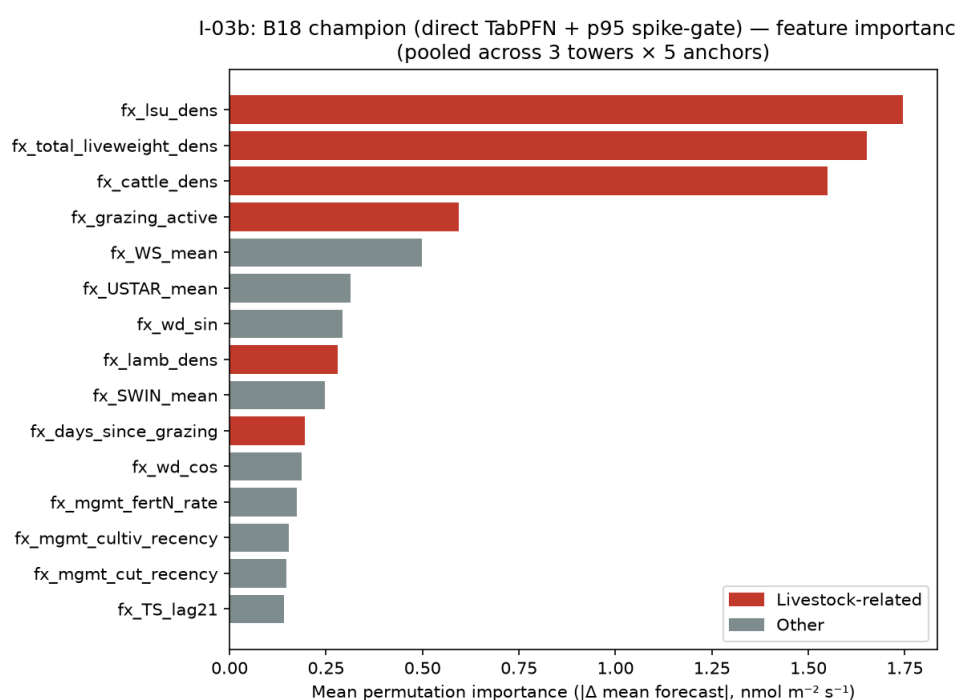


Figure 5.2: p95 event-corrected TabPFN champion (BASE\_ALL\_52): permutation feature importance, pooled across all three towers and five rollout anchors. Livestock-related features (red) dominate the top of the ranking.

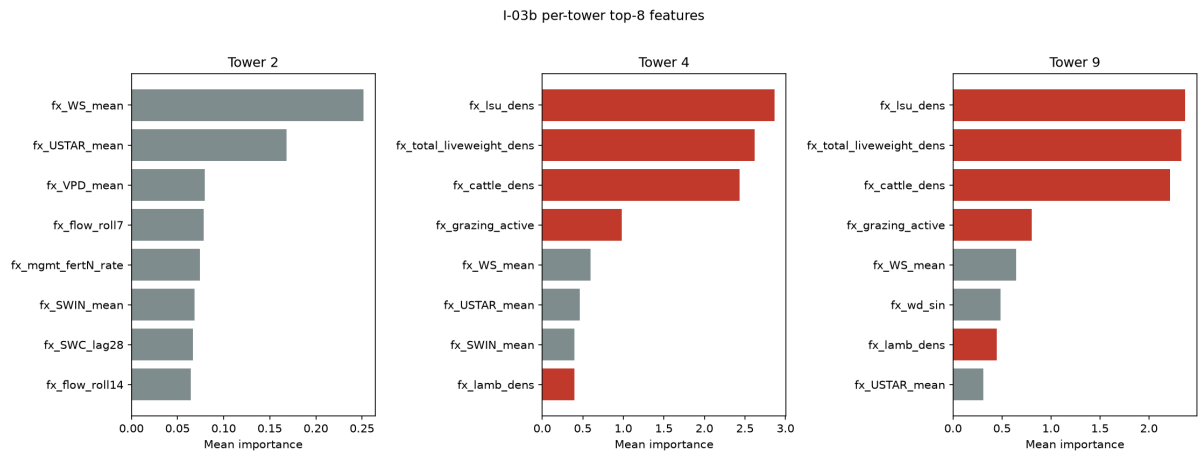


Figure 5.3: Per-tower top-8 permutation importance. Tower 2’s top features are exclusively meteorological, hydrological, and management-recency (wind speed, friction velocity, vapour-pressure deficit, catchment flow, fertiliser recency); Towers 4 and 9 both lead with livestock density and cattle density specifically.

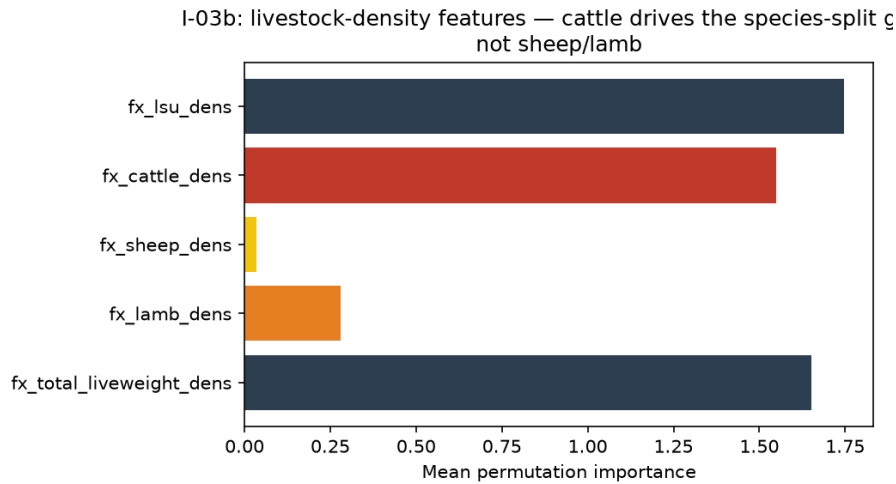


Figure 5.4: Species-disaggregated livestock-density importance: cattle density drives most of the species-split gain, though lamb density is a notable, unexplained exception (main text) and sheep density stays low.

One tower-level detail is worth reporting separately, with an explicit metric-convention warning: evaluated per-tower with MASE scaled against chain-*persistence* rather than seasonal mean (a different, non-interchangeable convention from every MASE figure elsewhere in this chapter, per Section 5.1.1), the earlier, un-refined TabPFN+species configuration achieves persistence-MASE of 0.229 at Tower 2, 0.883 at Tower 4, and 0.897 at Tower 9. Tower 2’s very low error magnitude here should not be read as a genuinely easier forecasting task: it reflects a much smaller and structurally different target distribution together with a small evaluation sample (102 rows), not better model tracking – consistent with Tower 2’s established livestock-blindness (this section, above) rather than any underlying ease.

### 5.3.5 Uncertainty Quantification

Uncertainty quantification in this chapter proceeds in two stages: a first, broad comparison across eight model families to establish general calibration behaviour, followed by a champion-focused recalibration on TabPFN and TabICLv2 specifically, and finally two targeted diagnostics – spike coverage and livestock-density-stratified calibration – on the champion pair. As throughout this dissertation, Area-of-Applicability is not used as an uncertainty-quantification method; PICP (Prediction Interval Coverage Probability) and MPIW (Mean Prediction Interval Width) are the primary uncertainty vocabulary, alongside pinball loss for the quantile-based comparisons below. Raw model quantiles are passed through leave-one-anchor-out conformal calibration – each of the five rollout anchors takes a turn as the calibration set for the other four – targeting 90% nominal coverage.

#### 5.3.5.1 All-Model Uncertainty Comparison

Raw calibration quality varies sharply by model family before any correction is applied: tree models (Random Forest, LightGBM, XGBoost) are badly overconfident (raw PICP

| Model                    | $n$  | Raw PICP | Raw MPIW | Raw pinball | Conf. PICP   | Conf. MPIW    | Conf. pinball |
|--------------------------|------|----------|----------|-------------|--------------|---------------|---------------|
| Ensemble (unweighted)    | 2217 | 0.732    | 78.82    | 10.47       | 0.890        | <b>145.28</b> | <b>10.53</b>  |
| Ensemble (MASE-weighted) | 2217 | 0.728    | 77.98    | 10.48       | 0.892        | 145.31        | <b>10.53</b>  |
| Random Forest            | 2217 | 0.379    | 47.18    | 12.23       | 0.886        | 145.77        | <b>10.53</b>  |
| LightGBM                 | 2217 | 0.489    | 55.22    | 11.38       | <b>0.894</b> | 149.79        | 10.70         |
| XGBoost                  | 2217 | 0.485    | 56.36    | 11.38       | 0.891        | 148.67        | 10.74         |
| SARIMAX                  | 2217 | 0.871    | 156.52   | 11.17       | 0.892        | 158.65        | 11.04         |
| TFT                      | 2217 | 0.839    | 145.63   | 10.92       | <b>0.894</b> | 162.89        | 11.49         |
| TabPFN                   | 2217 | 0.807    | 120.31   | 11.18       | 0.892        | 165.42        | 11.55         |

Table 5.8: First-stage uncertainty comparison, T4/T9 pooled test rows (Tower 2 lacked sufficient calibration support). Raw quantiles as produced by each model, and the same after leave-one-anchor-out conformal calibration.

0.38–0.49, far below the 0.90 target), while SARIMAX, TFT, and TabPFN start much closer to nominal (raw PICP 0.81–0.87). Conformal calibration equalises marginal coverage across every model family (0.886–0.894 after calibration), but achieves this primarily by widening the previously overconfident tree models’ intervals, not by narrowing anything materially. The ensembles and Random Forest achieve the lowest conformal pinball loss (10.53) in this first, broad comparison, marginally ahead of the champion-focused models examined next.

### 5.3.5.2 Conformal Calibration for Best Models

| Model                                 | Tower | $n$  | Raw PICP | Conf. PICP | Conf. MPIW   | Conf. pinball |
|---------------------------------------|-------|------|----------|------------|--------------|---------------|
| <b>TabPFN (B18 champion)</b>          | T4    | 1318 | 0.929    | 0.898      | <b>144.5</b> | <b>10.09</b>  |
| <b>TabPFN (B18 champion)</b>          | T9    | 899  | 0.928    | 0.894      | 166.4        | 11.42         |
| TabICLv2 (Ch5 champion)               | T4    | 1318 | 0.964    | 0.895      | 154.7        | 10.63         |
| TabICLv2 (Ch5 champion)               | T9    | 899  | 0.771    | 0.894      | 195.3        | 13.04         |
| TabICLv2 (scenario-safe) <sup>‡</sup> | T4    | 1318 | 0.959    | 0.895      | 151.3        | 10.71         |
| TabICLv2 (scenario-safe) <sup>‡</sup> | T9    | 704  | 0.991    | 0.884      | 161.2        | 11.47         |

Table 5.9: Champion-focused conformal recalibration. <sup>‡</sup>`Direct_TabICLv2_solo_trend` on the restricted `FX_A_SPECIES` feature set – the actual scenario-projection engine used in Chapter 6, distinct from this chapter’s own TabICLv2 champion configuration.

Recalibrating specifically on the champion configurations narrows the comparison to the models this dissertation actually recommends for forecasting (TabPFN, TabICLv2) and for scenario projection (the restricted-feature TabICLv2 variant above). Tower 2

cannot support calibration at all in any of these three passes – too few real observed values fall inside its rollout anchor windows to build a usable calibration set, the same structural limitation identified for Tower 2 throughout this dissertation (raw-only figures: TabPFN B18 champion PICP 0.726/MPIW 86.0,  $n = 102$ ; scenario-safe TabICLv2 PICP 0.804/MPIW 182.9,  $n = 102$ ). Recalibrated specifically for B18’s actual champion architecture (direct `BASE_ALL_52` regression plus the p95 spike-gate, rather than the earlier TS-wrapper configuration), TabPFN’s raw PICP now starts much closer to nominal at both towers (0.929/0.928, against 0.867/0.724 under the earlier configuration) – plausibly because the richer, 52-column feature set’s native quantile spread is better calibrated before any correction is applied. After conformal calibration, all three model/tower combinations converge close to the 0.90 target (0.884–0.898), as intended, and B18’s TabPFN champion is now the sharpest and best-scored of the three once calibrated (MPIW 144.5/166.4, pinball 10.09/11.42) – a modest but genuine improvement over both its own earlier-configuration numbers and TabICLv2’s. The scenario-safe TabICLv2 variant, despite its deliberately restricted 13-feature set, calibrates to essentially the same standard as the two full-feature champions (conf. PICP 0.895/0.884, MPIW 151.3/161.2) – reassuring given that this is the specific configuration Chapter 6’s scenario projections actually depend on.

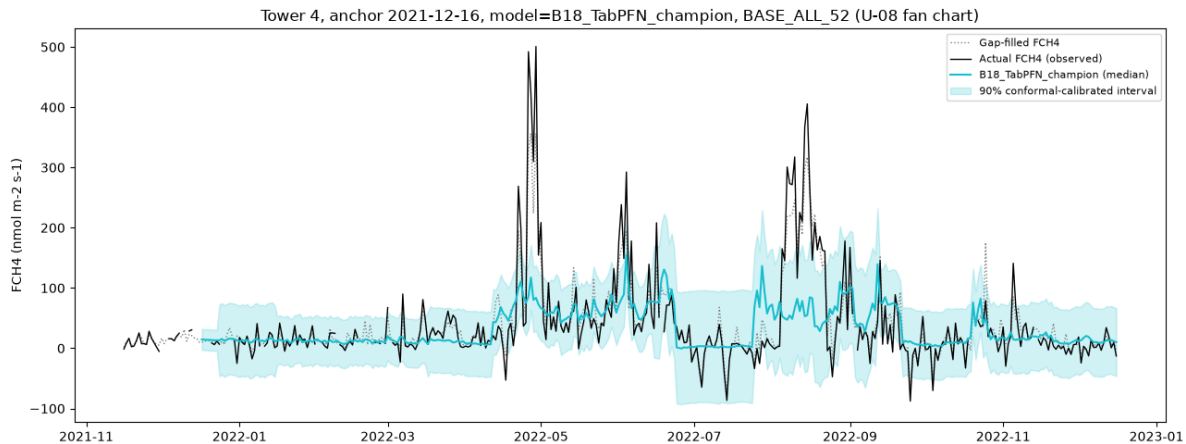


Figure 5.5: Fan chart, Tower 4, anchor December 2021, B18’s TabPFN champion (BASE\_ALL\_52): median forecast and 90% conformal-calibrated interval against observed and gap-filled FCH4. The interval widens visibly during the mid-year, high-livestock-density window and still does not fully contain every large spike – the motivation for the CQR spike-coverage correction below.

### 5.3.5.3 Spike-Coverage Correction (CQR)

The fan chart above motivates a direct question this dissertation can now answer quantitatively: does Conformalized Quantile Regression (CQR), applied specifically to widen intervals conditional on predicted spike risk, meaningfully close the spike-undercoverage gap identified throughout this chapter (Section 5.3.2)? Applying CQR to both champion-tier models’ saved chains gives a clear yes:

| Model                    | Spike coverage (raw) | Spike coverage (CQR) | Non-spike coverage (raw) | Non-spike coverage (CQR) |
|--------------------------|----------------------|----------------------|--------------------------|--------------------------|
| TabPFN (B18 champion)    | 0.266                | <b>0.775</b>         | 0.966                    | 0.879                    |
| TabICLv2 (scenario-safe) | 0.266                | <b>0.901</b>         | 0.961                    | 0.832                    |

Table 5.10: Spike-day interval coverage before and after CQR correction, both champion-tier models.

Both models’ raw, uncalibrated spike coverage sits at a near-identical 26.6% – markedly worse than the deep-dive TabPFN+species configuration’s own raw spike coverage reported in Section 5.3.2 (50.2%), a reminder that raw interval width alone is not a stable quantity across configurations. CQR correction roughly triples spike coverage for both

models (to 77.5% for the B18 champion and 90.1% for the scenario-safe TabICLv2 variant), at a real but modest cost to non-spike coverage ( $0.966 \rightarrow 0.879$  and  $0.961 \rightarrow 0.832$  respectively) – intervals widen selectively where spike risk is predicted to be high, trading a small amount of everyday-day coverage for a large gain exactly where this chapter’s central limitation (Section 5.3.2, Chapter 7) is sharpest. This meaningfully updates the picture from the deep-dive configuration’s own CQR result (spike coverage improved only to approximately 57%, Section 5.3.2): applied to either current champion-tier model, CQR now closes most, not merely some, of the nominal-coverage gap on the days that matter most for the CH<sub>4</sub> budget.

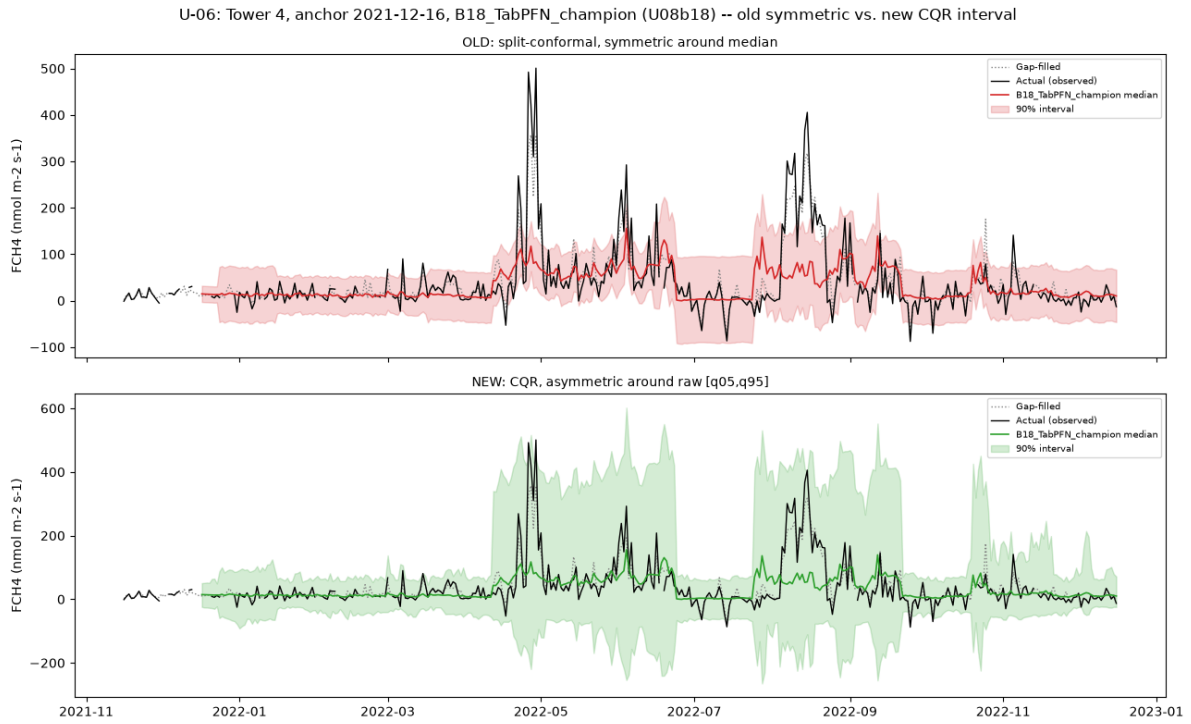


Figure 5.6: Tower 4, anchor December 2021, B18’s TabPFN champion: raw vs. CQR-corrected prediction intervals around observed and gap-filled FCH<sub>4</sub>, illustrating the selective widening that produces the spike-coverage gain in the table above.

The table above pools Towers 4 and 9 only, consistent with every conformal-calibration result in this chapter (Section 5.3.5.2) – Tower 2 cannot support the leave-one-anchor-out

calibration step at all. This is stated in prose throughout this chapter but has not, until now, been shown directly:

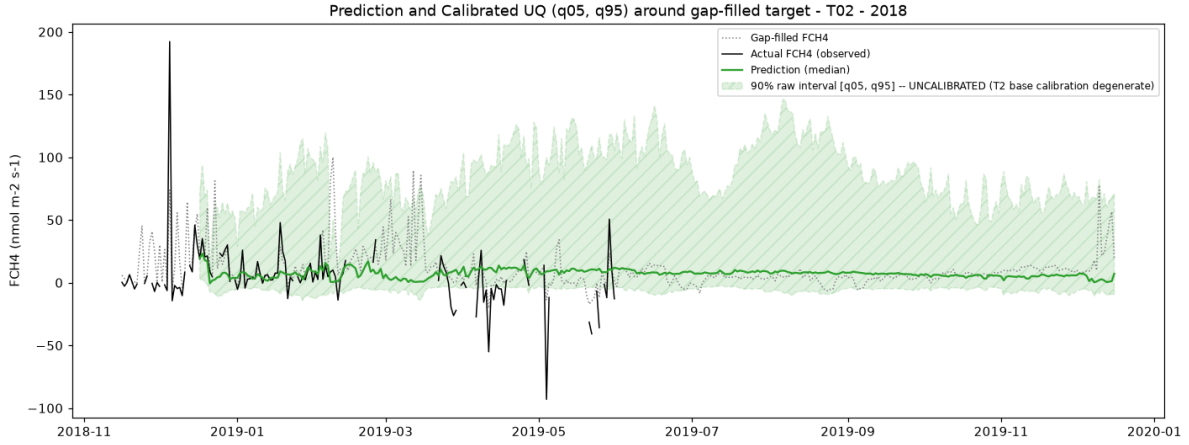


Figure 5.7: Tower 2, anchor December 2018 (its only anchor with real observed data, Section 3.2.1): the model’s raw, *uncalibrated*  $[q_{05}, q_{95}]$  interval, shown hatched to distinguish it from the CQR-calibrated bands at Towers 4 and 9 above. No CQR margin exists for Tower 2 at any anchor – this is what "Tower 2 cannot support calibration" actually looks like, not a narrower version of the calibrated case.

Both figures above are drawn from a full three-tower, five-anchor set (15 total figures, one per tower-anchor combination) generated for this dissertation’s own verification, consistent with this project’s standing full-coverage convention (Chapter 1); the two shown here are the representative examples, and the pooled Tower 4/9 table above already reflects all fifteen combinations, not just the anchor illustrated.

#### 5.3.5.4 Livestock-Conditional Uncertainty

Section 5.3.2’s spike-vs-non-spike split conditions on the observed *outcome*. A complementary, forward-looking stratification instead conditions on livestock density itself – an input known before the fact, motivated directly by this chapter’s own interpretability finding (Section 5.3.4) that livestock density is the dominant driver of FCH4:

Interval width and pinball loss both grow sharply with livestock-density tier for both

| Model                    | LSU tier | $n$ | PICP  | MPIW  | Pinball |
|--------------------------|----------|-----|-------|-------|---------|
| TabPFN (B18 champion)    | Low      | 855 | 0.885 | 91.4  | 4.73    |
| TabPFN (B18 champion)    | Mid      | 466 | 0.882 | 108.7 | 5.39    |
| TabPFN (B18 champion)    | High     | 875 | 0.825 | 305.2 | 17.32   |
| TabICLv2 (scenario-safe) | Low      | 811 | 0.836 | 166.8 | 6.53    |
| TabICLv2 (scenario-safe) | Mid      | 383 | 0.833 | 213.3 | 6.84    |
| TabICLv2 (scenario-safe) | High     | 807 | 0.865 | 381.8 | 17.37   |

Table 5.11: Livestock-density-stratified CQR coverage, width, and pinball loss, both champion-tier models.

models – roughly  $3.3\times$  wider at the high tier than the low tier for the B18 champion and  $2.3\times$  for the scenario-safe TabICLv2 variant – directly quantifying the regime-dependent uncertainty this chapter’s interpretability work already identifies as the dominant source of predictive difficulty. Coverage stays close to, and for the B18 champion’s low/mid tiers slightly above, the earlier deep-dive configuration’s own figures, remaining modestly below the nominal 0.90 target in every tier for both models (0.825–0.885) – a residual gap this dissertation reports honestly rather than tuning away, since narrowing it further would risk re-introducing the spike-undercoverage problem quantified in Section 5.3.2 and directly above.

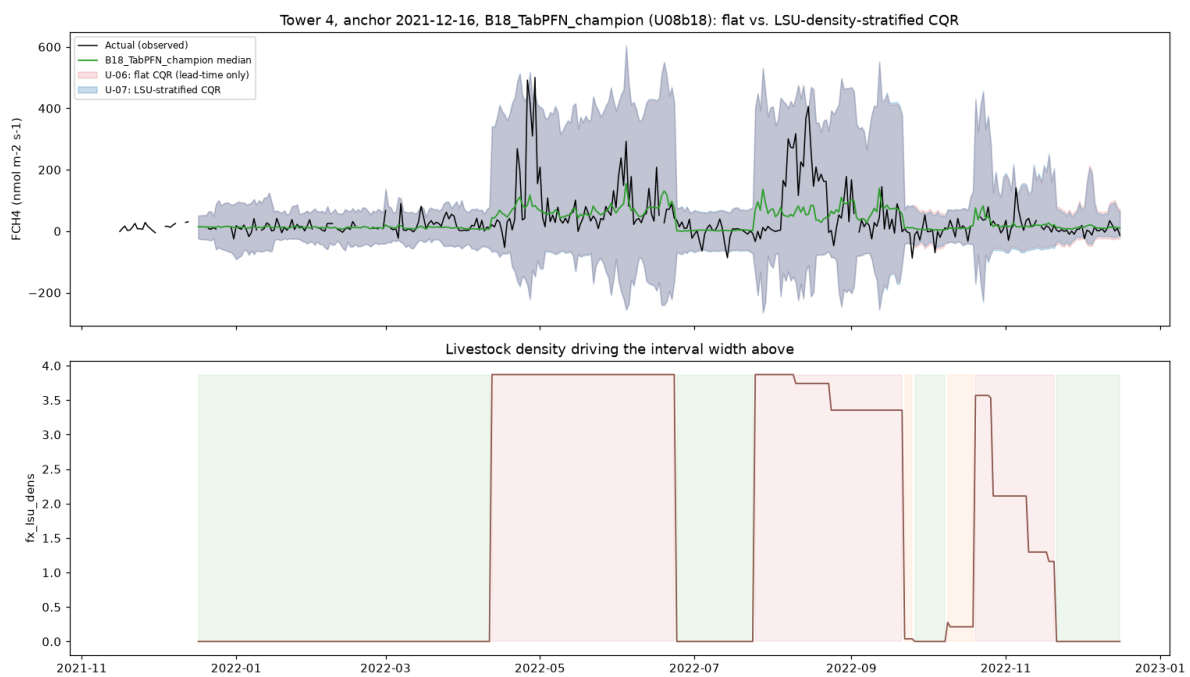


Figure 5.8: Tower 4, anchor December 2021: flat CQR (lead-time-only margin) vs. livestock-density-stratified CQR intervals, with the underlying  $fx\_lsu\_dens$  trajectory shown below – the stratified interval visibly widens exactly where livestock density is high, independent of which specific model produced the underlying point forecast.

## 6 | Scenario Projection

Scenario projection is structurally different from every chapter preceding it in this dissertation. Chapters 4 and 5 are both checkable against real observations – a gap-filled or forecast value can always, in principle, be compared against a genuine held-out FCH<sub>4</sub> measurement. Scenario projection has no such anchor: there is no ground truth for CH<sub>4</sub> flux under a livestock density, grazing pattern, or climate trajectory that has never actually occurred at NWFP, and there never will be, regardless of how the underlying model is built. Every result in this chapter should be read with that structural limitation in mind; it is returned to explicitly in Section 6.6.

### 6.1 Outline of Scenario Projection Task

Scenario projection extends the forecasting architecture of Chapter 5 to produce CH<sub>4</sub> projections under conditions that depart from NWFP’s observed historical record – altered livestock stocking density, shifted grazing timing, adjusted fertiliser application, and future climate trajectories drawn from CMIP6 projections – rather than forecasting forward from real, currently-observed conditions. This dissertation’s scenario architecture asks a single central "what-if" question along several axes: how does projected CH<sub>4</sub> respond to a change in livestock management or climate, holding everything else at its historically-observed pattern.

### 6.2 Interpretability and Feature Importance

The scenario architecture developed in this chapter is deliberately built around livestock density as its central management lever, not as an unmotivated design choice. This is grounded in the convergent interpretability evidence already established across gap-filling and forecasting (Chapter 5): livestock density is the single feature ranked most

important by every method capable of producing an attribution on this project’s model roster – native tree-model importance and SHAP (Random Forest, XGBoost, LightGBM), SARIMAX coefficients, and TabPFN permutation importance – at every tower except Tower 2, and species-disaggregated permutation importance on this dissertation’s forecasting champion further isolates that dominance to the cattle component specifically (Chapter 5). A dedicated permutation-importance pass on TabICLv2’s own one-shot output, specific to the simplified architecture this chapter uses (Section 6.4.3), had not yet been executed as of this writing – a natural, low-cost follow-up flagged here rather than assumed. The livestock-centric, species-disaggregated design of this chapter’s scenario architecture is accordingly grounded in convergent evidence from a different model and a different chapter’s work, not a fresh finding specific to this exact configuration.

## 6.3 Simulated Climate Dataset

Future climate projections are drawn from Semenov et al.’s large-ensemble dataset of transient CMIP6-based climate scenarios for Great Britain [22], providing daily weather series for the North Wyke location under multiple Shared Socioeconomic Pathways (SSPs) and multiple Global Climate Models (GCMs), each contributing several stochastic realisations. This dissertation uses SSP2-4.5 and SSP5-8.5, spanning five GCMs (ACCESS-ESM1-5, CNRM-CM6-1, HadGEM3-GC31-LL, MPI-ESM1-2-LR, MRI-ESM2-0), transient from 2020 through to a 2050 projection horizon.

## 6.4 Data Pre-processing and Model Adjustment

### 6.4.1 Standardisation of New Data with the NWFP Dataset

Two preprocessing steps standardise the CMIP6 series against NWFP’s own historical baseline before any scenario inference. First, of the roughly eleven driver variables this

dissertation’s forecasting feature set uses, four are derived directly or near-directly from the CMIP6 series itself (minimum, maximum, and mean air temperature; precipitation; and shortwave radiation, the latter requiring a verified unit conversion from CMIP6’s MJ/m<sup>2</sup>/day to this project’s W/m<sup>2</sup> convention); the remaining variables, for which CMIP6 provides no equivalent projection, are resampled from NWFP’s own historical day-of-year climatology. Second, a direct comparison of CMIP6’s own historical-period values against NWFP’s real observed climate found a genuine, non-trivial mismatch rather than a close match, in three distinct directions at once: raw simulated air temperature runs roughly 2–3.5°C too cool, raw simulated precipitation is roughly 4.3 times too wet, and raw simulated incoming shortwave radiation is roughly 8% too low, each relative to NWFP’s own real historical record. This motivated a delta/anomaly-style bias correction, applied per GCM: an additive offset for minimum and maximum air temperature, and a multiplicative ratio for shortwave radiation and precipitation, each calibrated against the difference between CMIP6’s own historical-period statistics and NWFP’s real observed record over the same period. The five correction factors used in this dissertation are summarised below.

| GCM             | Offset TA_min (°C) | Offset TA_max (°C) | Ratio SWIN | Ratio Precipitation |
|-----------------|--------------------|--------------------|------------|---------------------|
| ACCESS-ESM1-5   | 1.81               | 2.60               | 1.034      | 0.225               |
| CNRM-CM6-1      | 2.26               | 3.09               | 1.083      | 0.229               |
| HadGEM3-GC31-LL | 1.97               | 2.79               | 1.044      | 0.236               |
| MPI-ESM1-2-LR   | 2.71               | 3.56               | 1.060      | 0.226               |
| MRI-ESM2-0      | 2.29               | 3.03               | 1.043      | 0.240               |

Table 6.1: Per-GCM bias-correction factors applied to CMIP6 driver variables against NWFP’s own historical baseline.

Every result in Section 6.5 uses these bias-corrected drivers rather than CMIP6’s raw output. This is a materially different, and preferred, standardisation to the coarser, output-level bias correction used earlier in this project’s scenario-analysis development (correcting a model’s predicted output to match the real historical mean, rather than correcting its input drivers) – correcting the climate inputs themselves is the more defen-

sible fix where the underlying mismatch is demonstrably in the input data, rather than in the model’s own extrapolation behaviour.

## 6.4.2 Generation of Simplified Scenario Model

The scenario architecture used throughout Section 6.5 is `Direct_TabICLv2_solo_trend`: a per-tower direct tabular regression fit (`TabICLv2Regressor.fit/predict`, one fit per tower, reused across every scenario combination for that tower) on the restricted `FX_A_SPECIES` feature set plus a single linear trend covariate (`b17_days_since_2010`), with no pooling across towers and no spike-gate correction. This supersedes an earlier one-shot, no-fit in-context configuration (`TabICLForecaster`) used in this project’s initial scenario-engine development, following B18’s (Chapter 5) demonstration that direct-regression architectures generally outperform time-series-wrapper forecasting on this task: a real-anchor backtest (5 anchors, 3 towers, climatology-scored MASE, the same convention used throughout Chapter 5) found every direct-regression candidate beat the prior one-shot production architecture by a real margin, though the specific configuration mattered. A first, pooled-across-towers candidate beat the prior architecture by 4.4% MASE (0.726 vs. 0.760), but pooling assumes one shared training cutoff across all three towers, which does not match this project’s real per-tower anchoring pipeline; a faithful, solo-per-tower adaptation without pooling recovered only 0.6% of that margin (noise-level), until adding the trend covariate back in isolated the true driver and recovered most of the original gain (0.738 vs. 0.760, +**2.79% MASE**) – this is the locked-in `Direct_TabICLv2_solo_trend` configuration used throughout this chapter. The trend covariate’s own training range (2018–2025, roughly 2,557–5,480 days since 2010) is around  $2.7\times$  shorter than the horizon it is asked to extrapolate to at 2050 ( $\sim 14,600$  days) – a genuine extrapolation risk that this dissertation checked directly rather than assumed away: querying the fitted trend term 27–47 years past its own training maximum shows it saturates to a bounded, plausible value ( $\sim 8.3 \text{ nmol m}^{-2} \text{ s}^{-1}$ ) rather than diverging, unlike SARIMAX’s known

explosive extrapolation behaviour under a comparable stress test (Chapter 5). This is a single, targeted sanity check on the trend term specifically, not a full stress test of every scenario perturbation in Section 6.5 – a distinction returned to in Section 6.6.

This remains a deliberately narrower feature configuration than Chapter 5’s forecasting champion, using the same 13 `FX_A_SPECIES` features as before (mean/minimum/maximum air temperature, mean shortwave radiation, summed precipitation, day-of-year sine/cosine, growing-season and winter indicator flags, and species-disaggregated live-stock density – combined, cattle, sheep, and lamb) plus the trend term above, rather than the roughly 52-column enriched feature set Chapter 5’s champion uses. It remains important to be explicit that TabICLv2 is not this dissertation’s forecasting champion – that remains the TabPFN-based blend described in Chapter 5 (MASE=0.691) – TabICLv2 is selected here specifically because this simplified architecture’s smaller, CMIP6-derivable feature set is a better match to what a genuine future climate scenario can actually supply as covariates, and because the direct-regression family degrades more gracefully than the alternatives once restricted this way (the driver-availability ablation cited in Section 6.4.1 uses TabPFN’s own reduced-driver result as the motivating evidence for this family choice, not yet the final `FX_A_SPECIES` + trend configuration reported below).

### 6.4.3 Result of the Simplified Model

| Configuration                                                                                | MASE (climatology) |
|----------------------------------------------------------------------------------------------|--------------------|
| TabICLv2, BASE + ALL (Chapter 5 config)                                                      | 0.7385             |
| TabICLv2, BASE + species (Chapter 5 config)                                                  | 0.7405             |
| TabICLv2, <code>FX_A_SPECIES</code> , one-shot wrapper (superseded engine)                   | 0.7588             |
| TabICLv2, <code>FX_A_SPECIES</code> + fertiliser v2 <sup>†</sup>                             | 0.7591             |
| <b>Direct_TabICLv2_solo_trend, <code>FX_A_SPECIES</code> + trend (this chapter’s engine)</b> | <b>0.738</b>       |

Table 6.2: Cost of simplifying to a scenario-realistic feature set, TabICLv2 only.  $R^2$  is not reported for forecasting-style comparisons anywhere in this dissertation (Section 5.1.3).

<sup>†</sup>Fertiliser-variable extension tested only against the superseded one-shot architecture, not retested against the current engine.

Restricting the feature set to only what a genuine climate scenario can supply still carries

a small, quantifiable accuracy cost relative to TabICLv2’s own full-feature configurations from Chapter 5 (MASE rises from 0.7405, **BASE+species**, to 0.738 under this chapter’s current engine, an increase of around 0.3% – a smaller gap than the superseded one-shot architecture’s own 2.5% cost, since switching to direct regression recovers most of that difference) – a modest, reportable cost rather than a disqualifying one. A further fertiliser-variable extension (**FX\_A\_SPECIES + fertiliser v2**), tested against the earlier one-shot architecture, did not improve on it and was not adopted, consistent with this project’s general finding that fertiliser-related features add little once a richer base feature set is already present (Section 6.5.2); it has not been retested against the current **Direct\_TabICLv2\_solo\_trend** engine, and is not expected to change that conclusion given the null result already found on the superseded architecture.

## 6.5 Inference with Simulated Climate Dataset

### 6.5.1 Base Case

The scenario architecture’s baseline case holds livestock stocking density, grazing pattern, and fertiliser schedule at their historically-observed day-of-year patterns, varying only the underlying climate trajectory (SSP pathway, GCM, and realisation) drawn from the bias-corrected CMIP6 series described above. This baseline is the reference point every induced perturbation below is measured against.

### 6.5.2 Induced Perturbations

#### 6.5.2.1 Livestock Density

Livestock density is varied along a four-point ladder, externally anchored rather than built from an arbitrary multiplier: the historically-observed baseline ( $1\times$ ); half that density ( $0.5\times$ ); a literature-informed stocking ceiling for UK grassland (drawn from stewardship

stocking-density guidance); and each tower's own historical maximum observed density. Two variants are tested at each ladder rung: scaling all livestock species jointly, and scaling cattle alone (motivated by Section 6.2's finding that cattle specifically drives the livestock-density signal).

| Level                            | T2 (% vs. baseline) | T4 (% vs. baseline) | T9 (% vs. baseline) |
|----------------------------------|---------------------|---------------------|---------------------|
| Baseline (1×)                    | –                   | –                   | –                   |
| Half (0.5×), all species         | –27.1%              | –14.7%              | –27.0%              |
| Half (0.5×), cattle alone        | –24.6%              | –14.0%              | –17.6%              |
| Literature ceiling, all species  | +4.5%               | +0.9%               | –2.6%               |
| Literature ceiling, cattle alone | –2.1%               | +0.9%               | –2.6%               |
| Own historical max, all species  | +98.7%              | +137.8%             | +182.9%             |
| Own historical max, cattle alone | +57.6%              | +132.5%             | +187.6%             |

Table 6.3: Annual-mean FCH4 change relative to the historically-observed baseline, under the bias-corrected livestock-density ladder, this chapter's current `Direct_TabICLv2_solo_trend` scenario engine.

Each tower's own historical maximum livestock density sits in a different position relative to the literature ceiling used here, directly reflecting the differing management history described in Chapter 2: Tower 4 and Tower 9 have both, at points in their own record, exceeded the literature stocking ceiling used in this ladder, so their "own historical max" scenario represents a livestock density genuinely higher than the literature-informed reference point – consistent with NWFP's identity as an intensively-managed research platform rather than a typical extensive UK grassland site (Chapter 2). Tower 9's negative response at the literature-ceiling rung (–2.6%) is the clearest signature of this: scaling Tower 9's livestock *down* to the literature ceiling actually reduces projected FCH4 relative to baseline, because Tower 9's real historical stocking density already sits above that ceiling. Tower 2's own historical maximum, by contrast, never reached this literature ceiling, reflecting its distinct land-use history (the permanent-pasture-to-arable conversion described in Section 3.2.1). The literature-ceiling reference value used to produce the table above is 2.5 LSU/ha, drawn from UK Countryside Stewardship stocking-density

guidance (Annex 8, non-Severely-Disadvantaged-Area land).

The scaling response is not uniformly weaker at Tower 2 across every rung of this ladder: at the "half" and "literature-ceiling" rungs, Tower 2's proportional swing is comparable to, or larger than, Towers 4 and 9's – a consequence of Tower 2's much smaller absolute baseline ( $27.7 \text{ kg CH}_4 \text{ ha}^{-1} \text{ yr}^{-1}$ , against 72.4 and 92.2 at Towers 4 and 9 respectively, reported in full below), so the same absolute change in  $\text{kg ha}^{-1}$  terms produces a much larger percentage swing off a smaller base, not evidence of a stronger underlying livestock response. At the "own historical maximum" rung specifically – the most extreme, and the only rung anchored to each tower's own real management history rather than an external reference – Tower 2's response is decisively and consistently weaker than Towers 4 or 9's (+98.7% against +137.8% and +182.9%), consistent with Tower 2's established livestock-blindness (Chapter 4, Section 4.5.3.1; Chapter 5, Section 5.3.4) – a real model-behaviour finding independently confirmed here for a fourth time, under a different model architecture and different climate driver source than every prior confirmation, though this dissertation is explicit that the effect is rung-dependent rather than uniform across the whole ladder.

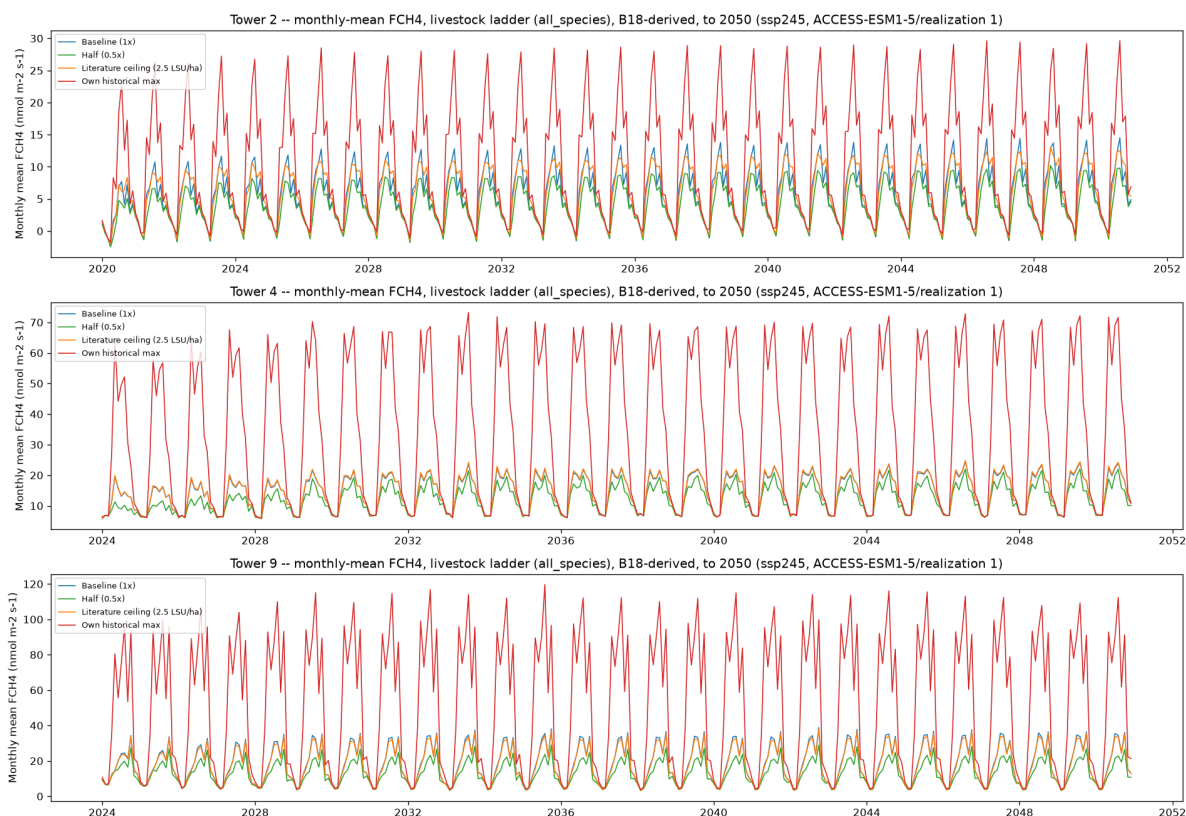


Figure 6.1: Monthly-mean projected FCH4 to 2050 under the livestock-density ladder (SSP2-4.5, one representative GCM/realisation), all three towers, this chapter's current scenario engine. Own-historical-max visibly separates from the other three rungs at every tower.

**Annual CH4 generation in absolute terms.** The percentage changes above are all relative to each tower’s own historical baseline, which differs sharply in absolute magnitude across towers and can make a small absolute change look large, or vice versa (the literature-ceiling discussion above is one direct example). The livestock-density ladder is reported below in absolute annual CH4 generation terms instead:

| Level                           | T2 (kg ha <sup>-1</sup> yr <sup>-1</sup> ) | T4 (kg ha <sup>-1</sup> yr <sup>-1</sup> ) | T9 (kg ha <sup>-1</sup> yr <sup>-1</sup> ) |
|---------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| Baseline (1×)                   | 27.7                                       | 72.4                                       | 92.2                                       |
| Half (0.5×), all species        | 20.2                                       | 61.7                                       | 67.3                                       |
| Literature ceiling, all species | 29.0                                       | 73.0                                       | 89.7                                       |
| Own historical max, all species | 55.1                                       | 172.1                                      | 260.7                                      |

Table 6.4: Projected annual CH4 generation (kg CH4 per hectare per year) under the livestock-density ladder, SSP2-4.5, this chapter’s current scenario engine.

In absolute terms, Tower 9’s baseline and own-historical-max generation both exceed Tower 4’s, which in turn exceeds Tower 2’s, at every rung of the ladder – the same tower ordering the percentage table above can obscure at the smaller rungs. Own-historical-max remains the largest single lever tested anywhere in this chapter in absolute terms too, reaching 260.7 kg ha<sup>-1</sup> yr<sup>-1</sup> at Tower 9, nearly three times its own baseline.

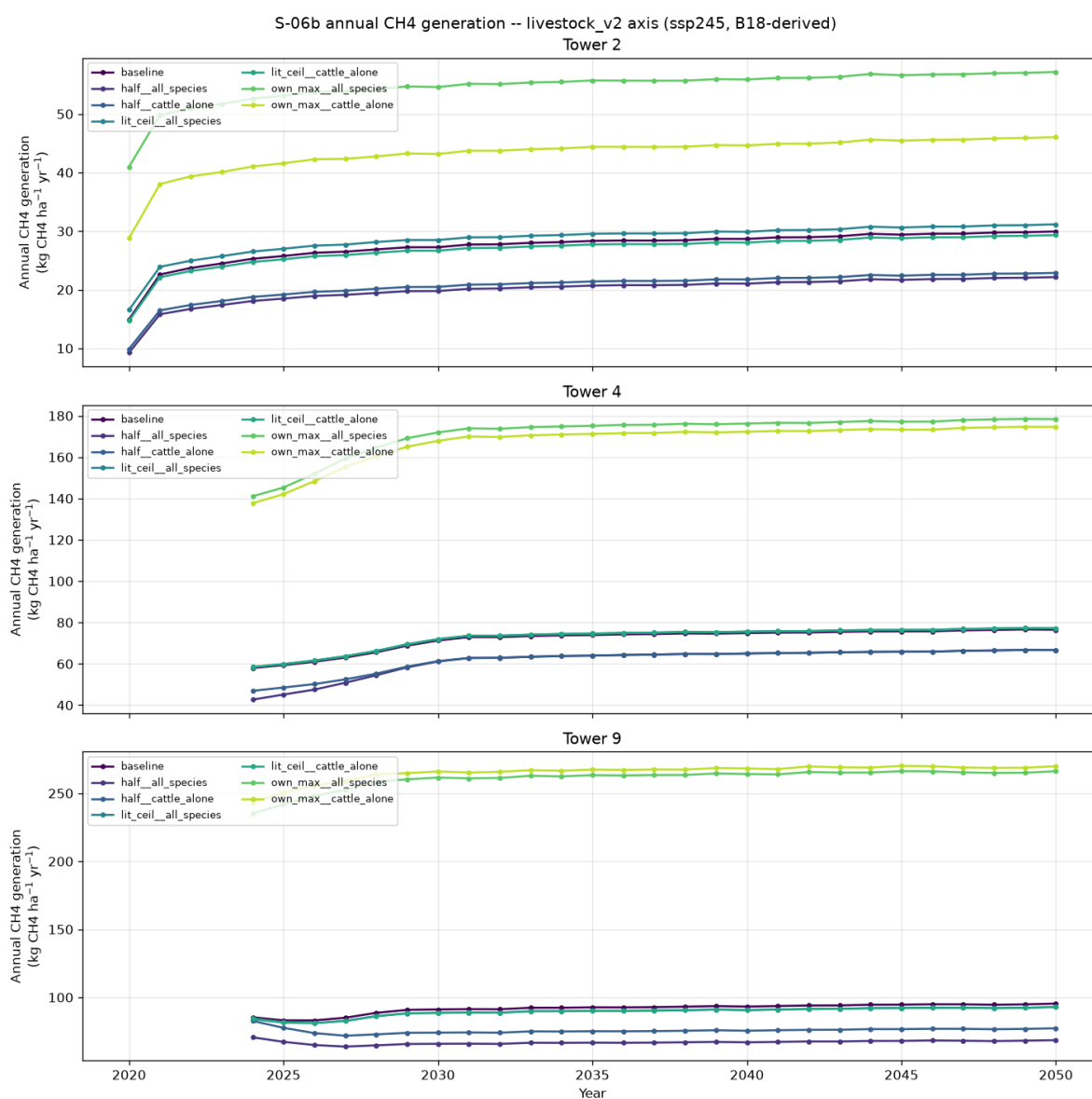


Figure 6.2: Projected annual CH<sub>4</sub> generation (kg CH<sub>4</sub> per hectare per year) under the livestock-density ladder, SSP2-4.5, all three towers. Grazing-timing and fertiliser-schedule perturbations (Section 6.5.2) produce the analogous absolute-terms figures, not reproduced here since their percentage effects are already small (Table above, Section 6.5.2).

**6.5.2.2 Grazing Timing and Fertiliser Schedule**

| Intervention                     | T2    | T4     | T9     |
|----------------------------------|-------|--------|--------|
| Extend grazing season by 4 weeks | −3.8% | +19.9% | +19.9% |
| Fertiliser frequency +50%        | −8.2% | −0.1%  | ≈ 0%   |
| Fertiliser rate +50%             | −1.7% | −2.0%  | ≈ 0%   |

Table 6.5: Annual-mean FCH4 change relative to baseline under grazing-timing and fertiliser perturbations (SSP2-4.5), this chapter’s current `Direct_TabICLv2_solo_trend` scenario engine.

Two further management perturbations are tested against the same bias-corrected climate driver pipeline. Shifting the grazing season four weeks later still produces a real, monotonic increase in projected FCH4 at Towers 4 and 9 (+19.9% each, materially larger than this same perturbation’s +18.6% under the earlier one-shot scenario engine, though the qualitative conclusion – a real, management-actionable lever – is unchanged). Tower 2’s response, however, *reverses sign* under this chapter’s current engine: −3.8%, against +3.8% under the earlier architecture. This dissertation reports this reversal directly rather than smoothing over it: Tower 2’s grazing response was already the smallest-magnitude and least livestock-driven result in this table under the earlier architecture, consistent with Tower 2’s established livestock-blindness (Chapter 4, Section 4.5.3.1; Chapter 5, Section 5.3.4), so a sign flip at this small a magnitude is consistent with Tower 2 genuinely lacking a stable livestock-driven signal to extend, rather than the underlying architecture change producing a materially different real-world finding – Towers 4 and 9’s own management-actionable conclusion is unaffected either way. Fertiliser schedule perturbations (a 50% increase in either application frequency or per-event rate) produce a small, still sign-inconsistent-across-towers effect, now reaching as much as −8.2% at Tower 2 (frequency) rather than the earlier architecture’s −3.5% – a larger magnitude than before, but still consistent with this project’s standing conclusion that fertiliser timing carries little independent signal once livestock density and the richer base feature set are already present, and still the smallest management lever tested in this chapter

for Towers 4 and 9 specifically ( $\leq 2.0\%$  in either direction). Separately, the SSP2-4.5 vs. SSP5-8.5 pathway divergence is small relative to every management perturbation above throughout most of the projection horizon, though it grows toward the end of the horizon and reaches its largest value at Tower 2 (approximately 2.9% divergence late in the horizon) – a minor lever next to the livestock and grazing-timing effects reported above, but not strictly negligible everywhere.

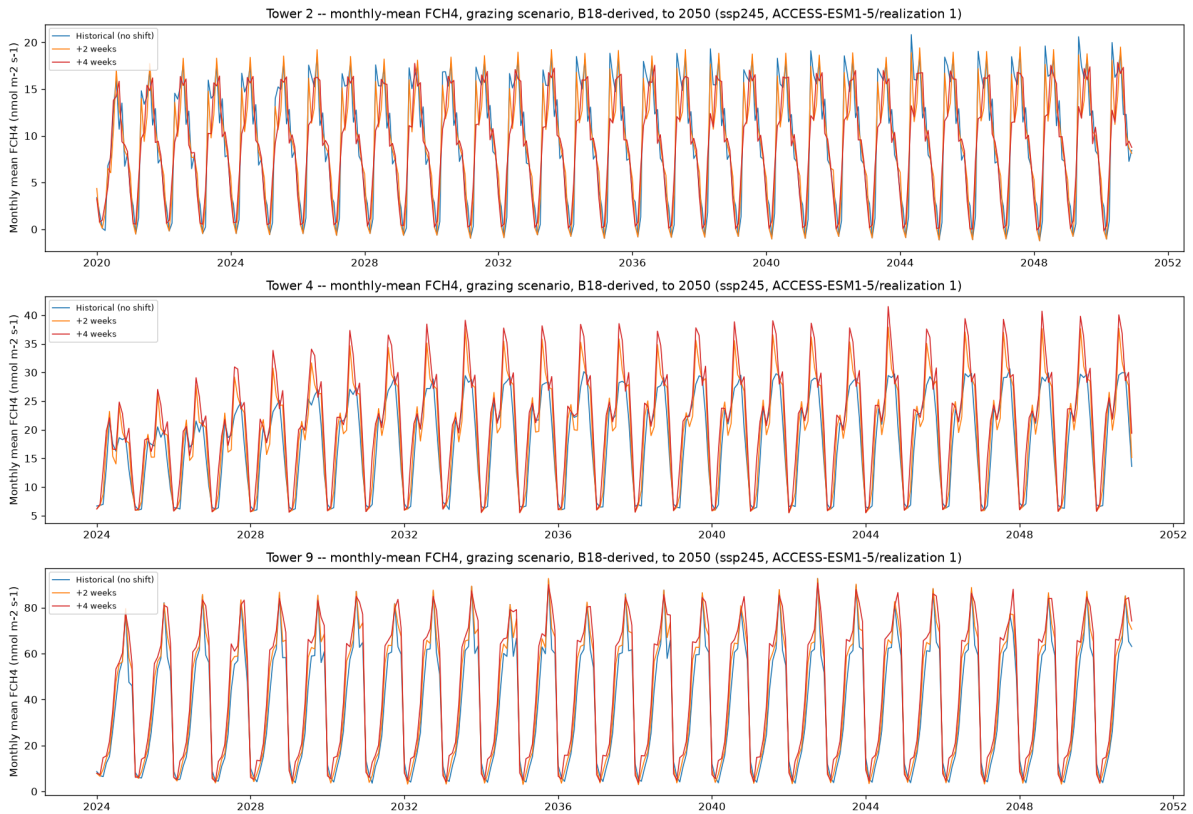


Figure 6.3: Monthly-mean projected FCH<sub>4</sub> to 2050 under the grazing-timing shift (SSP2-4.5, one representative GCM/realisation), all three towers, this chapter's current scenario engine. The shifted-timing curve sits visibly above the historical-timing curve at Towers 4 and 9, and is slightly below it at Tower 2.

## 6.6 Limitations

Three limitations apply specifically to this chapter’s scenario projections, beyond the general absence of ground truth stated in its opening framing.

First, TabICLv2’s extrapolation behaviour under genuine distribution shift has not been formally stress-tested to the standard Chapter 5’s uncertainty-quantification stress test applied to the production ensemble – TabICLv2 was not part of that model roster, and this chapter’s livestock-ladder sweep, while exposing the model to out-of-historical-range livestock density, is a behavioural probe rather than a controlled extrapolation test with any ground truth to check against. TabICLv2 belongs to the TabPFN-family of in-context tabular models, a family with a documented structural extrapolation limitation under genuine distribution shift (Section 2.4). One narrower piece of this concern was checked directly rather than left as an open risk: the current engine’s trend covariate (Section 6.4.3) is fitted on roughly 2018–2025 data and queried out to 2050, a genuine extrapolation in time, and a direct sanity check found it saturates to a bounded, plausible value rather than diverging when queried decades past its own training range. This is reassuring for the time-trend extrapolation specifically, but it is a single, targeted check on one covariate, not a substitute for a full stress test of the model’s behaviour under livestock-density or climate-driver distribution shift – the two forms of extrapolation this chapter’s scenarios actually rely on most heavily. This remains a limitation of this chapter’s headline result specifically, not a secondary footnote, since the livestock-density and climate scenarios reported above are precisely the kind of distribution shift this limitation concerns.

Second, no diagnostic available to this dissertation can detect a smooth-but-confidently-wrong extrapolation. This is a structural gap rather than a tooling gap that could simply be built: historical PICP (Chapter 5) measures calibration only at real, historically-anchored points, and cannot be computed at all for a genuine future scenario, since no

future ground truth exists to check against; a narrow prediction interval width is fully compatible with a confidently wrong extrapolation; and GCM/realisation spread (Section 6.3) measures sensitivity to the climate input alone, not error in the underlying model’s own extrapolation. One further distinction is stated explicitly here to avoid it being misread elsewhere in this dissertation: prediction intervals were successfully generated for over 99% of eligible scenario rows in this chapter’s pipeline, but this is *interval availability* (the pipeline ran without failure), not *empirical coverage* – it says nothing about whether those intervals are actually well calibrated under scenario conditions, for exactly the reason given above.

Third, the CMIP6-vs-NWFP historical mismatch identified in Section 6.4.1 is a real, separate limitation independent of model choice – it reflects a genuine discrepancy between the global climate model ensemble’s own historical baseline and NWFP’s real local climate record, corrected for as best this dissertation can (Section 6.4.1), but not eliminated as a source of residual uncertainty in the corrected driver series feeding every result in Section 6.5.

## 7.1 Main Findings

### 7.1.1 Overview of Model Performance

Across both gap-filling (Chapter 4) and forecasting (Chapter 5), the same broad ranking recurs: tabular foundation models (TabICLv2, TabPFN) lead, tree-ensemble and statistical models follow closely behind, and deep sequence models (Bi-LSTM, DLinear, TFT on its own base feature set) trail well behind both. This is best understood as two separate, already-established mechanisms stacked together, not one blanket explanation.

The first mechanism is specific to this dissertation’s data: **missingness-compensation**. FCH4 target missingness at NWFP is severe – 56.5% at Tower 4, 75.0% at Tower 9, and 88.5% at Tower 2 (Section 3.2.1) – and every parameter-fitting model family (Random Forest, XGBoost, LSTM, SARIMAX) must learn its parameters directly from whatever clean signal survives that missingness. TabPFN and TabICLv2, by contrast, are pre-trained on large volumes of synthetic tabular data and make predictions via in-context learning at inference time, fitting no dataset-specific parameters to NWFP’s own record at all (Section 2.4). This structurally insulates them from a problem that measurably hurts every other model family here, and is the reason heavy missingness is treated as a thesis-wide theme in this dissertation rather than a detail confined to Chapter 4.

The second mechanism is not specific to this dissertation at all: Zeng et al.’s finding, already introduced in Chapter 2, that permutation-invariant self-attention destroys the temporal ordering information sequence models depend on, causing sophisticated attention-based architectures to lose to much simpler tree-based and even linear baselines on standard long-horizon forecasting benchmarks. Chapter 5’s own results substantiate this directly: DLinear and LSTM perform far worse than every tree, statistical, or foun-

dation model tested, even at each model’s own best-found configuration (all-tower MASE of 1.187 and 0.952 respectively, against every other model’s MASE below 0.88, Section 5.3.1), and TFT only reaches the mid-pack ranking in Table 5.1 once given the enriched feature set the tree and ensemble models gained nothing from (Section 5.2) – reinforcing that architectural sophistication cannot be assumed to translate into forecasting accuracy on this task.

Two scope notes are worth stating precisely rather than left implicit. In gap-filling, TabICLv2 leads at two of three towers (Tower 2 and Tower 4), not all three – RFm edges it narrowly at Tower 9 (Section 4.5.3) – so this dissertation states "leads at 2/3 towers," not an unqualified "wins." In forecasting, TabPFN’s win is unambiguous across the full model roster and is not qualified in the same way. Separately, this ranking is stated purely on raw benchmark performance: whether a configuration has matching production-grade tooling built around it (Section 4.5.3) is irrelevant to the ranking itself – this dissertation reports benchmark numbers, not a deployment recommendation.

### 7.1.2 Difficulty in Methane Flux Gap-Filling and Forecasting

Even at this dissertation’s best-achieved results, absolute performance stays modest by the standard of many other applied ML domains: gap-filling’s champion configuration reaches hourly  $R^2_{OLS}$  of 0.41–0.60 (Chapter 4), and forecasting’s champion, while beating a defensible climatology baseline by a wide margin (MASE=0.691), does not do so uniformly – Section 5.3.2’s spike-vs-non-spike evaluation shows that advantage essentially disappears on high-magnitude days (Chapter 5). This is not a failure of the modelling approach so much as a reflection of the difficulty already established in Chapter 2: CH<sub>4</sub> flux at a managed grassland is driven by the continuous interaction of non-stationary biophysical drivers together with discrete, spike-inducing livestock management events, evaluated here against a published literature ceiling of  $R^2 < 0.1$  at hourly resolution for the same ecosystem type – a ceiling this dissertation’s gap-filling result clears by

a wide margin, even as the absolute numbers remain modest in isolation. Both this dissertation’s results and the wider literature reviewed in Chapter 2 point to the same underlying source of that difficulty: a genuinely spike-dominated, high-variance target, not a deficiency specific to any one model family tested here.

## 7.2 Limitations

**Data quality and reliance on artificial (gap-filled) data.** Forecasting’s autoregressive history is seeded from Chapter 4’s gap-filled series rather than exclusively real observations, and this dissertation’s own results show that choice is not neutral: scoring against the gap-filled target instead of real observations reverses the model ranking entirely (Section 5.1.3), a direct consequence of forecasting’s weaker performance being at least partly rooted in the quality and completeness of the underlying gap-filled data it depends on. Tower 2 is the clearest single illustration of this limitation across the whole dissertation: its severe missingness, its distinct land-use-conversion history, its  $R^2$ -comparability caveat (Section 4.5.3.1), and its inability to support conformal calibration specifically in the forecasting rollout (Section 5.3.5.2 – note this is a forecasting-rollout-specific limitation, not a gap-filling one: Tower 2’s gap-filling uncertainty intervals, Section 4.5.4, are unaffected and calibrate normally) are each already established in full elsewhere in this dissertation; the summary point here is that Tower 2 is consistently this project’s worst-conditioned data source, not a re-telling of why.

**Scenario projection as a pure inference task.** As stated at the opening of Chapter 6, no future ground truth exists to validate a scenario projection against, and no existing diagnostic in this dissertation substitutes for that absence: PICP and MPIW (Chapter 5) measure interval calibration at real, historically-anchored points, not trustworthiness under genuine distribution shift, and nothing in the current pipeline can catch a smooth, confident, but structurally wrong extrapolation (Section 6.6).

**CMIP6-vs-NWFP historical climate mismatch.** Independent of model choice, the CMIP6 climate projections underlying Chapter 6’s scenario work do not, in their raw form, match NWFP’s own observed historical climate (Section 6.4.1); a delta/anomaly-style correction is applied, but does not eliminate this as a residual source of uncertainty in every downstream scenario result.

**Compute-constrained uncertainty-quantification and scenario scope.** Uncertainty quantification and scenario projection were built specifically for the models suited to each architecture’s own requirements – one-shot inference, no fitting step, vectorised full-trajectory prediction – and within the compute budget available to this project, not restricted to only the best-performing models by some other criterion. This distinction matters because TabICLv2, the model underlying Chapter 6’s scenario architecture, is not this dissertation’s forecasting champion (that remains TabPFN with species features, Chapter 5); TabICLv2 was chosen for its fit to scenario inference’s specific structural requirements, not because it is the most accurate forecaster available.

## 7.3 Future Work

Two extensions follow directly from findings established in this dissertation, rather than being generic placeholders. First, **livestock geolocation tracking relative to each tower’s footprint** – for example via satellite/remote-sensing platforms such as Google Earth Engine – could directly explain spike occurrence rather than relying on catchment-level stocking density as a proxy, since this dissertation’s interpretability findings (Chapter 5, Chapter 6) consistently identify livestock presence as the dominant CH<sub>4</sub> driver without yet being able to resolve exactly when and where within a catchment that presence translates into a flux spike; this has the potential to improve both gap-filling and forecasting performance directly. Second, **rerunning this dissertation’s full pipeline as further NWFP data accumulates** is a genuinely actionable, low-effort extension

rather than a generic "more data would help" statement: NWFP's data collection is ongoing and continuous, and every model, feature, and evaluation script built in this dissertation is designed to accept an extended record without structural change.

## 8.1 Summary and Contribution

This dissertation set out, against the five objectives stated in Chapter 1, to develop and evaluate machine learning approaches for CH<sub>4</sub> forecasting at the North Wyke Farm Platform and to demonstrate their integration with scenario projection under contrasting management and climate conditions. Each objective has been met directly: a systematic review of the relevant literature (Chapter 2) established that ecosystem-scale CH<sub>4</sub> forecasting at a managed temperate grassland was, to the author’s knowledge, previously unaddressed; NWFP’s EC CH<sub>4</sub> record was acquired, corrected, and gap-filled to a continuous series (Chapter 3, Chapter 4); a benchmarking pipeline spanning statistical baselines, tree ensembles, and tabular foundation models was designed and evaluated under temporal cross-validation throughout (Chapter 4, Chapter 5); interpretability and uncertainty quantification were applied across every stage (Chapter 4, Chapter 5, Chapter 6); and synthetic management and climate scenarios were generated and evaluated against the resulting models (Chapter 6).

The research questions posed in Chapter 1 are each answered directly by this work. RQ1’s survey of ML approaches for CH<sub>4</sub> gap-filling and its distinction from multi-step forecasting is answered in Chapter 2: gap-filling and forecasting are structurally different problems, and forecasting at this ecosystem scale had no prior treatment in the literature. RQ2’s comparison of statistical baselines, tree ensembles, and deep learning under NWFP’s own temporal variability is answered in Chapters 4 and 5: tabular foundation models lead, tree ensembles follow closely, and deep sequence models trail, for the two separable reasons established in Chapter 7. RQ3’s question of transferability from adjacent domains is answered throughout Chapter 4: some published findings (Random Forest beating MDS, ANN’s competitiveness at longer gaps) replicate cleanly at NWFP,

while others (PCA degrading ML performance) do not, underscoring that findings from wetland and rice-paddy sites do not transfer uniformly to managed UK grassland. RQ4's question of how interpretability and uncertainty quantification decode the interaction between continuous environmental drivers and discrete management events is answered by this dissertation's single most robust finding, discussed below. RQ5's question of the structural requirements for integrating forecasting into scenario-based "what-if" analysis is answered by Chapter 6's working scenario architecture and, equally importantly, by the structural limitations that architecture surfaces (Section 7.2) – what such a system requires is now considerably better understood than it was at the outset of this project, including where it currently falls short.

If this dissertation is remembered for a single finding, it should be this one: **livestock density is the dominant driver of CH<sub>4</sub> flux at NWFP, and this is not a fragile, single-method result.** It recurs independently across gap-filling feature importance, forecasting SHAP attribution and permutation importance, SARIMAX regression coefficients, and scenario projection's own livestock-ladder response – genuinely different methods, different model architectures, and different chapters of this dissertation, each arriving at the same answer independently. Most empirical work produces one headline number and hopes it survives scrutiny; this project instead produced a finding that shows up no matter which door is used to go looking for it.

Alongside the modelling results, this dissertation's evaluation-methodology corrections deserve explicit credit rather than being read as small next to the headline model numbers, since several of them were larger in effect than any single feature-engineering gain in the entire project. Tower 2's evaluation-protocol fix alone (Section 4.4.3) moved its gap-filling sklearn  $R^2$  from  $-16.9$  under an early, seasonally-mismatched train/test split to a positive result under the corrected full-period gap-CV protocol used throughout Chapter 4's improved pipeline (Section 4.5.3.1) – a bigger single correction than any feature added across the whole gap-filling programme. The reversal from single-anchor

to multi-anchor rollout evaluation is a genuine, non-obvious methodological lesson in its own right, not a footnote: single-anchor rankings in this project’s own preliminary work proved unreliable and would have supported different conclusions had they been trusted. And catching the observed-versus-gap-filled-target scoring circularity (Section 5.1.3) before it was allowed to corrupt a headline forecasting claim is exactly the kind of methodological rigour that is easy to skip and expensive to get wrong.

Finally, it is worth being precise about what this dissertation has, and has not, shown. On gap-filling, this project has demonstrated a significant, benchmarked improvement over traditional methods such as MDS and plain RFm, achieved through a combination of tabular foundation models and a substantially richer feature set than the published baselines this work replicates. On forecasting, this project has generated models that reliably beat naive and climatological baselines, but does not yet produce fully trustworthy point forecasts, particularly during the high-magnitude spike events that dominate CH<sub>4</sub>’s error budget throughout this dissertation. Scenario projection, in turn, demonstrates that the resulting architecture can produce distinguishable, interpretable projections under contrasting management and climate conditions, while carrying the structural limitation that no future scenario can ever be checked against ground truth. Taken together, this dissertation delivers a benchmarked, uncertainty-quantified, and interpretable foundation for CH<sub>4</sub> forecasting and scenario projection at a managed UK grassland site that did not previously exist in the literature – not a finished, deployment-ready forecasting system, but a substantial and honestly-bounded step toward one.

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## A | Source Code

Source code for all of the methods implemented in Chapters 3-6 for the project can be found in the GitHub repository:

[link](#).

